Protection and Dilution: SAFEs and MFN Clauses in Seed Financing

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Abstract

We ask how an accelerator should protect its stake against the seed investor who finances its startups next, and what a most-favored-nation (MFN) clause, such as the $375,000 uncapped MFN SAFE that Y Combinator added to its standard deal in 2022, is worth to it. In our model, the accelerator spends its contribution before a liquidity shock, and a seed investor with market power covers the rest of the shock with a priced or an unpriced SAFE. Without protection, the seed investor takes all of the equity of the projects it finances, except with an uncapped SAFE without discount, which it never uses: the accelerator keeps value only on the shocks it covers alone. A fixed post-money share gives the accelerator this share of every project the seed investor finances. When the optimal fixed share comes with a contribution, an MFN component does strictly better, whether the seed investor uses a priced or an unpriced SAFE: it takes more from small rounds than from large ones. Against the clause, the seed investor prefers the unpriced SAFE, and answers by investing beyond its need in the smallest rounds. Whatever the protection, a seed round covers at most the project's expected surplus at the scaling cost, which is below the scaling cost. Numerically, the accelerator invests, and puts almost all of its money through the clause, only when liquidity needs are small relative to the scaling cost. For larger needs, it invests nothing, and its protection, a fixed share of a project it does not fund, lowers the total surplus to 0.72–0.77 of the first best, which the absence of protection reaches.

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1 Introduction

Most accelerators offer the same standard deal to every startup in a batch. The most common deal is a small amount for a share of equity, which Techstars offered from its first program in 2007.[1] Y Combinator (YC) put on average about $17,000 in equity for 7% and $80,000 in a safe converting at the terms of the next round.[2] In 2014, it replaced this structure by a single deal, "$120k for 7% equity (regardless of the number of founders)", to avoid complexity.[3] Sam Altman, president of YC in 2014, argued: "Speaking of hating complexity, we've tried to make the new structure really simple. The convertible notes and safes we used got complicated in terms of how they got priced, and complexity often causes unintended consequences. It was hard for founders to actually predict how much total dilution they were looking at." But, in January 2022, YC added to its $125,000 for 7% a $375,000 uncapped Simple Agreement for Future Equity (SAFE) with a most-favored-nation (MFN) clause: the SAFE converts at the best terms the startup grants to intermediate investors before its first equity round. The deal is still the same for every startup in the batch, and according to Geoff Ralston, president of YC in 2022, it should "remove the immediate pressure to fundraise and accept less than favorable terms".[4] Three quarters of YC's money now go through the clause.

The MFN clause answers a structural problem of sequential venture financing. The accelerator sources and trains startups and hands them to venture capitalists (VCs) (Gonzalez-Uribe and Leatherbee, 2018; Hallen et al., 2020). When it invests, neither the startup's probability of success nor its exact financing need is known, an uncertainty that comes with early-stage financing (Nanda and Rhodes-Kropf, 2017; Lerner and Nanda, 2020; Ewens et al., 2018). Information arrives in sequence: the financing need is known before the quality (Kerr et al., 2014; Ewens et al., 2018; Bergemann and Hege, 1998, 2005). A startup from an accelerator may therefore need additional funding before it can raise an equity round (Coyle and Green, 2014), and the intermediate investors who provide it have market power over the entrepreneur. The terms they set dilute the accelerator, and a protection written before the need is known decides how much of the surplus the accelerator keeps. Nondiscrimination clauses such as MFN clauses are meant to protect early signatories against this opportunism, but McAfee and Schwartz (1994) show that they generally fail to do so. Ralston also announced that the new deal should help founders "raise a successful seed round".[5] This paper asks which protection an accelerator should write against the seed investor, within

[1] "Initially, Techstars invested between $6,000 and $18,000 in early stage companies" (Techstars, Wikipedia).

[2] TechCrunch, April 22, 2014.

[3] Sam Altman, "The New Deal," Y Combinator blog, April 22, 2014, which is also the source of the quotation below.

[4] "YC's $500,000 Standard Deal," Y Combinator blog, January 11, 2022, `https://www.ycombinator.com/blog/ycs-500-000-standard-deal`.

[5] Same source. The MFN SAFE "will take on the terms of the lowest cap safe (or other most favorable terms)" issued "between the start of the batch and the next equity round".

the family of SAFEs that accelerators and seed investors use, what an MFN clause is worth to it, and how the clause changes the total surplus, which we use to measure welfare.

We summarize our findings. First, without protection, the seed investor takes all of the equity of the projects it finances. The accelerator's contribution is spent before the shock, and the accelerator keeps value only on the shocks it covers alone (Propositions 1 and 3). The seed investor obtains the same with a capped SAFE or with an uncapped SAFE whose discount it chooses. An uncapped SAFE without discount pays back at most its amount, and the seed investor never uses it (Proposition 2). Second, a fixed post-money share gives the accelerator this share of every project the seed investor finances, and the accelerator earns strictly more than without protection (Propositions 4–6). Priced and unpriced SAFEs give the same outcome (Proposition 5). A capped SAFE for the accelerator does no better than the fixed share. It does as well when the fixed share keeps more than what the SAFE returns of its amount, since the seed investor invests only its need (Proposition 7). Third, when the optimal fixed share comes with a contribution, an MFN component strictly beats any fixed share, whether the seed investor uses an unpriced SAFE (Proposition 10) or a priced one (Proposition 11). The clause takes a share that falls with the round: on a small round, the accelerator's money weighs a lot against the seed investor's. Against the clause, the seed investor prefers the unpriced SAFE, which converts at the same price as the MFN component in every state (Proposition 8). It answers by investing beyond its need in the smallest rounds, and every dollar it adds is lost on the projects that are not fully funded. The fixed share survives only to curb this answer. Fourth, a seed round covers at most the project's expected surplus at the scaling cost, which is itself below the scaling cost (Proposition 12). A protection leaves the accelerator a share of the last project the seed investor finances, and the seed investor stops earlier (Proposition 13). A protection changes the probabilities of a seed round and of full funding through the contribution, which moves the seed investor's zone, and through what the accelerator keeps on the last project, which shortens it; the seed investor's investment beyond its need raises the probability of full funding (Propositions 14 and 15). The total surplus is maximal when the accelerator contributes nothing, and without protection it falls with the contribution. The protections move it either way, and the investment beyond the need lowers it (Propositions 17–19). Fifth, numerically, the two regimes separate sharply (Section 8). When liquidity needs are small relative to the scaling cost, $E(\ell)/I \leq 0.05$, the accelerator invests, almost all of its money goes through the clause, and the MFN clause gives the highest profit and the highest total surplus. Whether the fixed part of the clause is a share or the accelerator's own SAFE makes no difference. From $E(\ell)/I = 0.15$ on, the accelerator invests nothing under any protection: its protection is then a fixed share of a project it does not fund, and lowers the total surplus to 0.72–0.77 of the first best, which the absence of protection reaches.

To arrive at these conclusions, we set up a three-date game between risk-neutral agents. We see entrepreneurship as experimentation (Kerr et al., 2014): a project

must survive a liquidity shock before its value is revealed, and is worth funding only once the value is known. The entrepreneur sells the whole project to the accelerator, which spends a fixed amount with a protection before the shock. The seed investor observes the shock and makes a take-it-or-leave-it offer to cover it, with a priced or an unpriced SAFE. Once the value is revealed, a competitive market funds the projects worth funding through equity rounds. Money invested in the project never returns to its investors, and a rejected seed round abandons the project: the accelerator accepts any round. We solve the game by backward induction, first without protection (Section 3), then with a fixed post-money share or a capped SAFE for the accelerator (Section 4), then with an MFN component (Section 5), derive the probabilities of a seed round and of full funding (Section 6), and the total surplus (Section 7), and check the results against another distribution of the project's value and a stake kept by the founders (Section 9). All the analytical results hold for any log-concave distributions of the project's value and of the liquidity shock. Section 8 computes the optimal contracts for software and deep-tech projects.

Our contribution is twofold. First, we propose a tractable model of the accelerator's protection against a seed investor with market power, in which the protection is a cap on the share the seed investor may take, possibly depending on the amount it invests. Fixed shares and MFN clauses are special cases. To our knowledge, it is the first model of an accelerator's protection against seed dilution. Second, we evaluate the MFN clause that accelerators adopted after YC. The clause is worth more to the accelerator than any fixed share because it discriminates between rounds, and its value is limited by the seed investor's answer and by the accelerator's own contribution, which caps the amount the clause covers. Its effect on the total surplus is not signed in general: numerically, the clause raises it for software-like projects, and any protection lowers it for deep-tech projects.

Our paper relates to four literatures.

The first studies entrepreneurship as experimentation. Kerr et al. (2014) argue that the probability of success of a new venture is unknown until someone invests in it, and that the cost of experimenting shapes who finances innovation. Ewens et al. (2018) show that when cloud computing lowered this cost, VCs started to give little money and little governance to many more startups, and to abandon more of them. Nanda and Rhodes-Kropf (2017) show that investors can protect a startup against a lack of future funding by giving it more money upfront, at the cost of the option to stop. Investors also stage their financing to learn before investing more. Gompers (1995) shows that VCs stage more often when agency costs are higher. Neher (1999) explains staging by the entrepreneur's ability to renegotiate once the money is invested, early rounds building the collateral that supports later ones. Bergemann and Hege (1998, 2005) show that funding stops too early when the learning process is subject to moral hazard. We add to this literature by modeling the terms on which the experiment's cost is financed after the first investor has invested, and how the first investor protects itself.

The second studies the securities used in venture capital. Cornelli and Yosha (2003) show that convertible debt deters the entrepreneur from manipulating interim results to avoid liquidation, because a better interim result also makes the VC convert. Schmidt (2003) shows that convertible securities give both the entrepreneur and the VC incentives to invest in the project. Repullo and Suarez (2004) derive the optimal securities when the project is financed in two stages under double-sided moral hazard. In Casamatta (2003), the VC both finances and advises the firm, and receives common stock when it invests little and convertible securities when it invests a lot. Hellmann (1998) explains why entrepreneurs give VCs the right to replace them. Admati and Pfleiderer (1994) show that a VC who receives a fixed fraction of the payoff and finances the same fraction of every later round has no incentive to misprice later rounds. Kaplan and Strömberg (2003, 2004) document that VC contracts allocate cash-flow, board, voting and liquidation rights separately, and that riskier deals give the VC more of these rights. Coyle and Green (2014) document that after 2005, convertible notes, until then used for bridge loans, became a seed financing instrument with a conversion price cap, followed by the SAFE. We add to this literature by comparing a priced and an unpriced SAFE when an accelerator has invested before the seed investor. The two contracts give the same outcome against a fixed share, which caps what an unpriced SAFE takes when the next equity round values the project below its cap. Against an MFN clause, the seed investor prefers the unpriced SAFE, which converts at the same price as the accelerator's MFN component.

The third studies most-favored-customer clauses. Cooper (1986) shows that a firm that promises to rebate its past customers if it later cuts its price has less reason to cut it, so the clause helps duopolists keep prices high. Besanko and Lyon (1993) show that a contemporaneous clause, which only covers customers served at the same time, is a weaker device, and that not adopting it can be a dominant strategy. McAfee and Schwartz (1994) show that when a supplier contracts with competing buyers, nondiscrimination clauses generally do not protect a buyer against the supplier's renegotiation with the others, which can explain why franchise contracts are uniform. In these papers the clause protects a buyer. We add to this literature by studying a clause held by an investor against a later investor. Like the clause of Cooper (1986), YC's MFN SAFE gives an early signatory the better terms granted later. It is a price discrimination device: it takes more from the later investor when that investor's round is small.

The fourth studies the investors who finance a startup before VCs. Hellmann and Thiele (2015) model startups that raise angel money before VC money. The two investors are friends because each relies on the other's investment, and foes because VCs no longer need the angels at the later stage. Hellmann and Puri (2002) show that VC-backed startups professionalize faster, and Hochberg et al. (2007) show that the startups of better-networked VCs are more likely to raise a next round. On accelerators, Gonzalez-Uribe and Leatherbee (2018) find with Start-Up Chile that entrepreneurship schooling improves venture performance, while funding and coworking space alone do not. Hallen et al. (2020) find that some, but not all, accelerators

speed up venture development. Kim and Wagman (2014) model an accelerator that chooses its portfolio size and which of its signals to disclose, and Zarei et al. (2022) model the accelerator's selection, mentorship and investment decisions. These papers do not study the terms of the accelerator's investment. We add to this literature by modeling the accelerator's contract and its protection against the seed investor who follows.

2 Model

Projects. A project has two sunk costs: a liquidity shock $\ell \geq 0$, with density g and distribution function G, unknown ex ante; and a scaling cost $I > 0$, known ex ante. Its valuation $\pi \in [0, 1]$, with density f and distribution function F, is unknown ex ante. We write

$$W(x) \equiv \int_x^1 (\pi - x)\, f(\pi)\, d\pi,$$

with $W'(x) = -(1 - F(x))$, and $W(x) = E(\pi) - x$ for $x \leq 0$.

Assumptions.

(A1) $E(\pi) < I$.
(A2) f and g are positive and log-concave.
(A3) When all information is available, all NPV-positive projects obtain funding from a competitive market through equity rounds.
(A4) If ℓ is not covered, π is not revealed.

By (A1), no one funds a project before its value is known: the project must first be an experiment. Since $(\pi - I)^+ \leq \pi$, $W(I) \leq E(\pi)$, and (A1) also gives $W(I) < I$.

Accelerator. The entrepreneur initially sells all of the project to the accelerator, which invests σ_a on each project before ℓ and π are known, with a protection described below. The project spends this money: it never returns to the accelerator.

Seed investor. If $\ell > \sigma_a$, a seed investor invests σ_s after ℓ and before π, in exchange for $\rho_s N_t$ shares, where N_t is the number of shares available at date t.

(B1) The seed investor proposes take-it-or-leave-it offers.

An equity round that occurs at date t sets a price p_t per share and a number N_t of shares to transfer an amount σ in exchange for ρN_t shares. If an equity round has already occurred, $N_t > N_{t-1}$. We denote by $V_t = N_t p_t = \sigma/\rho$ the post-money valuation, and by $N_{t-1} p_t$ the pre-money valuation. If the accelerator's round is an equity round, N_0 is set at $t = 0$; otherwise, N_0 is set by the first equity round.

Funding of the project. Let $x = I + \ell - \sigma_a - \sigma_s$ be the threshold valuation above which the project is fully funded. By (A3), once π is revealed, a competitive investor funds x for the share x/π when $\pi > x$, in an equity round with pre-money valuation $\pi - x$, and the project is abandoned otherwise. An abandoned project returns nothing: the accelerator and the seed investor lose all the money they invested, including the cash beyond the need. The earlier shareholders hold equity worth $W(x)$ in expectation. If $\ell \leq \sigma_a$, the accelerator's cash covers the shock and the threshold is $I + \ell - \sigma_a$.

Contracts. The seed investor offers either a priced SAFE, which fixes its share ρ_s when it invests,[6] or an unpriced SAFE with cap $V_s \geq \sigma_s$ and discount $\lambda \in (0, 1]$, which converts at this equity round at the lower of V_s and $\lambda(\pi - x)$: it claims the share $\sigma_s / \min(V_s, \lambda(\pi - x))$ of the equity held before the round. The accelerator protects its stake by a cap on the share the seed investor may take. We study three protections:

- a fixed post-money share ρ_a: the seed investor's share may not exceed $1 - \rho_a$;
- a capped SAFE for the accelerator: its contribution claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$ of the equity, with cap $V_a \geq \sigma_a$ and discount $\lambda_a \in (0, 1]$;
- an MFN clause: the accelerator keeps at least ρ_a^0, and an amount $m \leq \sigma_a$ converts on the seed investor's terms, at the lower of its valuation and $\pi - x$ for a priced SAFE, and with its cap and discount for an unpriced SAFE.

The accelerator keeps the residual share. Its protection comes first, as in Y Combinator's post-money SAFE, under which "the safes are not diluted by each other":[7] the SAFEs signed after it share the equity it leaves. When they claim more than this equity, each keeps the same fraction of its claim.

Participation. If the accelerator rejects the seed round, the project is abandoned, and the accelerator, whose contribution is spent, receives nothing. It therefore accepts any round. The seed investor invests only if it earns a nonnegative profit.

Timing. At $t = 0$, the accelerator invests σ_a with its protection. The shock ℓ is then realized. At $t = 1$, the seed investor offers (ρ_s, σ_s), and the accelerator accepts or rejects. The value π is then revealed, and at $t = 2$ the project is fully funded if $\pi > x$.

[6] In this framework, an equity round that sells the share ρ_s is equivalent to a priced SAFE.

[7] Y Combinator, Post-Money Safe User Guide. The dilution from later SAFEs falls on the founders, and the standard SAFE does not say what happens when they have no equity left. The only pro rata rule of the SAFE applies when the proceeds of a liquidity or dissolution event fall short of the amounts due.

3 Benchmark: no protection

Proposition 1. *Without protection, for $\ell > \sigma_a$, the seed investor invests if and only if $\ell - \sigma_a \leq W(I)$. It then invests $\sigma_s^* = \ell - \sigma_a$, takes $\rho_s^* = 1$, and earns $W(I) - (\ell - \sigma_a)$; the accelerator keeps nothing.*

→ Proof in Appendix A, page 27

The accelerator's contribution, already spent, reduces the seed investor's need: the seed investor finances shocks up to $\sigma_a + W(I)$, and takes the whole project.

The seed investor may also use an unpriced SAFE. Without protection, the SAFE takes its claim as long as it does not exceed the whole equity: with cap V_s and discount λ, it receives $\min\big(\sigma_s / \min(V_s, \lambda(\pi - x)),\, 1\big)(\pi - x)$ in state $\pi > x$.

Proposition 2. *Without protection, for $\ell > \sigma_a$:*

(i) with an uncapped SAFE without discount, the seed investor loses $\int_0^{\sigma_s} F(x + t)\, dt > 0$ at every amount $\sigma_s > 0$, and finances no shock;

(ii) with an uncapped SAFE whose discount it chooses, or with a SAFE whose cap and discount it chooses, it obtains what it obtains with a priced SAFE (Proposition 1), on the same projects.

In both cases, the accelerator keeps nothing on the shocks $\ell > \sigma_a$.

→ Proof in Appendix A, page 27

An uncapped SAFE without discount receives $\min(\sigma_s, \pi - x)$: it is a debt of face value σ_s, which pays back at most its amount. A discount raises the face value to σ_s/λ, and a low enough discount, or a cap below the amount, gives the seed investor the whole equity.

The accelerator therefore keeps value only on the projects it finances alone, and solves

$$\max_{\sigma_a \geq 0} B(\sigma_a) = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

Proposition 3. *Every optimal contribution σ_a^B is zero or satisfies*

$$W(I)\, g(\sigma_a^B) = \big(1 - G(\sigma_a^B)\big) + \int_0^{\sigma_a^B} F(I + \ell - \sigma_a^B)\, g(\ell)\, d\ell.$$

→ Proof in Appendix A, page 28

The left side is the gain on the project $\ell = \sigma_a$, which moves from the seed investor, who takes all of it, to the accelerator. The right side is the loss of the extra dollar: on the shocks above the contribution, it goes to the seed investor or is lost on an abandoned project, and on the projects the accelerator finances alone, it is returned only when the project is fully funded.

4 A fixed post-money share

4.1 The seed investor

Proposition 4 (Priced SAFE)**.** *With a fixed share $\rho_a > 0$, for $\ell > \sigma_a$, the seed investor invests $\sigma_s^* = \ell - \sigma_a$, the threshold is I, and it takes $1 - \rho_a$. It earns $(1 - \rho_a)W(I) - (\ell - \sigma_a)$, and invests if and only if $\ell \leq \sigma_a + (1 - \rho_a)W(I)$; the accelerator keeps $\rho_a W(I)$.*

→ Proof in Appendix A, page 28

Proposition 5 (Unpriced SAFE)**.** *For $\ell > \sigma_a$, the seed investor obtains with an unpriced SAFE exactly what it obtains with a priced SAFE, on the same projects, and the discount plays no role. The cap $V_s^* = (\ell - \sigma_a)/(1 - \rho_a)$ gives it the share $1 - \rho_a$ in every state above I.*

→ Proof in Appendix A, page 29

An unpriced SAFE converts at the lower of its cap and the discounted pre-money valuation of the next equity round: when this round values the project below the cap, the SAFE returns at least its amount. The fixed share caps what the seed investor takes in these states, and a cap low enough gives it the share $1 - \rho_a$ in every state, as a priced SAFE does.

4.2 The accelerator

With a fixed share, the accelerator's profit is

$$\Pi_a(\sigma_a, \rho_a) = B(\sigma_a) + \rho_a W(I)\Big[G\big(\sigma_a + (1 - \rho_a)W(I)\big) - G(\sigma_a)\Big]$$

against a priced or an unpriced SAFE.

Proposition 6. *(i) At each σ_a, the optimal share $\rho_a^*(\sigma_a)$ is unique, lies in $(0, 1)$, and satisfies*

$$G\big(\sigma_a + (1 - \rho_a)W(I)\big) - G(\sigma_a) = \rho_a W(I)\, g\big(\sigma_a + (1 - \rho_a)W(I)\big).$$

(ii) Every optimal contribution σ_a^* is zero or satisfies

$$(1 - \rho_a^*)W(I)\, g(\sigma_a^*) + \rho_a^* W(I)\, g\big(\sigma_a^* + (1 - \rho_a^*)W(I)\big) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

(iii) The accelerator earns strictly more than in the benchmark.

→ Proof in Appendix A, page 29

A dollar more moves the whole zone of the seed investor by one dollar. The project $\ell = \sigma_a$ moves from the seed investor to the accelerator, which gains the seed investor's share of it, $(1-\rho_a)W(I)$, and a project enters at $\sigma_a + (1-\rho_a)W(I)$, where the accelerator keeps $\rho_a W(I)$. At a given contribution, the marginal value of the contribution differs from the benchmark by $\rho_a W(I)[g(\sigma_a + (1-\rho_a)W(I)) - g(\sigma_a)]$: the fixed share pushes the contribution up when the shock density is higher at the last project than at the first, and down otherwise.

4.3 A capped SAFE for the accelerator

Y Combinator's \$125,000 for 7% is itself a post-money SAFE.[8] With its own SAFE, the accelerator claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$ of the equity. Multiplied by $\pi - x$, this claim is the larger of the share σ_a/V_a of the equity, when the cap binds, and of the value σ_a/λ_a, when the discount binds. The protection takes at most the whole equity, and leaves in expectation

$$W\Big(x + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a} W\Big(x + \frac{V_a}{\lambda_a}\Big).$$

Proposition 7 (The accelerator's SAFE)**.** *With a capped SAFE for the accelerator, for $\ell > \sigma_a$, the seed investor invests $\sigma_s^* = \ell - \sigma_a$ and takes all the equity the protection leaves, with a priced ($\rho_s = 1$) or an unpriced SAFE. It invests if and only if*

$$\ell - \sigma_a \leq W\Big(I + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a} W\Big(I + \frac{V_a}{\lambda_a}\Big),$$

and the accelerator keeps $W(I)$ minus this value, at least $W(I) - W(I + \sigma_a)$, on each of these projects. The accelerator never earns more than with the fixed share. It earns as much when an optimal fixed share (σ_a^P, ρ_a^P) has $\sigma_a^P > 0$ and $\rho_a^P W(I) > W(I) - W(I + \sigma_a^P)$, and comes arbitrarily close to it when $\sigma_a^P = 0$.

→ Proof in Appendix A, page 30

The seed investor invests only its need, so what the protection takes below the scaling cost does not matter, and the accelerator's profit depends only on what it keeps at the threshold I, as with the fixed share. The SAFE only bounds this value from below: it keeps at least $W(I) - W(I + \sigma_a)$, the expected value of $\min(\sigma_a, (\pi - I)^+)$, since it returns its amount when the project allows it. In our simulations, the optimal fixed share keeps more than this bound whenever the accelerator invests (Section 8): capping well is then pricing well.

[8] "We invest \$125,000 on a post-money safe in return for 7% of your company" (The Y Combinator Deal).

5 The MFN clause

With an MFN clause, the accelerator keeps at least ρ_a^0, and its amount m converts on the seed investor's terms. Against a priced SAFE of valuation σ_s/ρ_s, its amount converts at $\min(\sigma_s/\rho_s, \pi - x)$. Against an unpriced SAFE, the two SAFEs convert at the same price, $\min(V_s, \lambda(\pi - x))$. For $m = 0$, the clause is a fixed post-money share.

5.1 The seed investor

Proposition 8 (The seed investor's contract)**.** *For $\ell > \sigma_a$, an unpriced SAFE gives the seed investor at least as much as a priced SAFE, and strictly more when $m > 0$. With an unpriced SAFE, the clause caps the seed investor's share at $(1-\rho_a^0)\sigma_s/(\sigma_s + m)$, which the cap $(\sigma_s + m)/(1-\rho_a^0)$ attains, and a discount brings it nothing. The seed investor finances the shock if and only if $\ell - \sigma_a + m \le (1-\rho_a^0)W(I)$, and invests beyond its need if*

$$(1-\rho_a^0)\Big[\frac{m\,W(I)}{(\ell-\sigma_a+m)^2} + \frac{(\ell-\sigma_a)\big(1-F(I)\big)}{\ell-\sigma_a+m}\Big] > 1.$$

→ Proof in Appendix A, page 31

A priced SAFE keeps its share when the equity round values the project below the seed investor's valuation, while the MFN component converts at the price of this round. The two claims then exceed the equity left by the fixed share, and both lose the same fraction of their claims. An unpriced SAFE converts at the same price as the MFN component in every state, and keeps the share $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ in all of them.

Every dollar the seed investor adds raises its share $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$: it dilutes the MFN component. The gain is large on small rounds, where m weighs a lot against σ_s.

Proposition 9 (Overinvestment in closed form)**.** *The condition of Proposition 8 holds exactly when $\ell - \sigma_a$ is below the positive root of the quadratic obtained by multiplying the condition of Proposition 8 by $(\ell-\sigma_a+m)^2$, which depends on f only through $F(I)$ and $W(I)$. On a round where it invests beyond its need, its amount at threshold x is the positive root*

$$\sigma_s = \frac{1}{2\big[1-(1-\rho_a^0)(1-F(x))\big]}\Bigg(\sqrt{\begin{aligned}&(1-\rho_a^0)^2(1-F(x))^2m^2\\ &+4\big[1-(1-\rho_a^0)(1-F(x))\big](1-\rho_a^0)\,m\,W(x)\end{aligned}} - m\big[2-(1-\rho_a^0)(1-F(x))\big]\Bigg),$$

the corresponding shock is $\ell - \sigma_a = \sigma_s - (I - x)$*, and the accelerator keeps* $\big(1 - (1 - \rho_a^0)\sigma_s/(\sigma_s + m)\big)W(x)$*.*

→ Proof in Appendix A, page 34

5.2 The accelerator

On a financed project, the accelerator keeps the project's value minus the seed investor's amount and profit. For $\theta = \rho_a^0$ or m, this value has derivative

$$-\frac{\partial \Pi_s}{\partial \theta}(\sigma_s^*) - \left[F(x^*) + \frac{\partial \Pi_s}{\partial \sigma_s}(\sigma_s^*)\right]\frac{\partial \sigma_s^*}{\partial \theta},$$

where Π_s is the seed investor's profit: what the clause takes from the seed investor at a given amount, minus the cost of the change in its amount, of which $F(x^*)$ per dollar is lost on the projects that are not fully funded.

Proposition 10. *Suppose that the optimal fixed share comes with a positive contribution, or that* $g(0) > 0$*. Every optimum* $(\sigma_a^*, \rho_a^{0*}, m^*)$ *satisfies:*

(i) $m^* > 0$*: an MFN component does strictly better than any fixed share;*
(ii) the seed investor invests beyond its need on the smallest rounds;
(iii) if $\rho_a^{0*} > 0$ *and* $m^* < \sigma_a^*$*, the gain from moving protection from the fixed share to the MFN component at a given last financed project, which is strictly positive on every round without overinvestment, equals its cost on the rounds with overinvestment, where every extra dollar is lost on the projects that are not fully funded.*

The accelerator earns strictly more than with a fixed share, hence than in the benchmark.

→ Proof in Appendix A, page 34

Point (iii) explains the fixed share. At a given last financed project, the MFN component takes the more from a round the further the round is from the last one: it is a price discrimination device. If the seed investor did not answer, the accelerator would move all of its protection into the clause, until $\rho_a^0 = 0$ or $m = \sigma_a$. The fixed share survives only to curb the seed investor's investment beyond its need, which a stronger fixed protection reduces. The clause covers at most the accelerator's contribution: when the accelerator invests nothing, the MFN component is zero, and the clause is a fixed share.

The seed investor prefers the unpriced SAFE (Proposition 8). If the accelerator could require a priced SAFE, the MFN clause would still beat any fixed share.

Proposition 11. *Suppose that the optimal fixed share comes with a positive contribution. Against a seed investor that uses a priced SAFE, every optimum has* $m^* > 0$*, and the accelerator earns strictly more than with any fixed share.*

→ Proof in Appendix A, page 38

The seed investor's share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$, so a small MFN component raises the value the accelerator keeps on every round by at least $(1-\rho_a^0)\,m\,W(I)/(\ell-\sigma_a+m)$, and the seed investor stops financing only rounds within $O(m)$ of the ends of its zone.

The fixed part of the clause can also be the accelerator's own SAFE, on the rest of its contribution, $\sigma_a - m$, as in Y Combinator's deal. The seed investor and the MFN component then share what this SAFE leaves, and the fixed share is the limit of a SAFE of cap $V_a = (\sigma_a - m)/\rho_a^0$ without discount as m tends to σ_a. The value $(\sigma_a - m)/\lambda_a$ that the SAFE keeps in the low states changes the seed investor's answer only on the small rounds where it invests beyond its need. Section 8 finds that it is worth nothing to the accelerator.

6 Seed rounds and full funding

The protection changes which projects survive. The probabilities of a seed round and of full funding describe this change; Section 7 measures welfare. The experiment is covered if the accelerator's cash covers the shock, $\ell \le \sigma_a$, or if a seed investor finances it. The project is fully funded if the experiment is covered and π exceeds the threshold. Its probability of full funding is

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell + \int_S \big(1 - F(x^*)\big) g(\ell)\, d\ell,$$

where S is the set of shocks the seed investor finances and x^* the threshold it chooses. A project the accelerator finances alone faces the threshold $I + \ell - \sigma_a \le I$, since the unused cash goes to the project.

Proposition 12. *Whatever the accelerator's protection and the seed contract, a seed investor finances a shock $\ell > \sigma_a$ only if $\ell - \sigma_a \le W(I)$. A seed round covers at most $W(I) < I$, and a project is covered only if $\ell \le \sigma_a + W(I)$.*

→ Proof in Appendix A, page 39

At best, the seed investor takes the whole equity, and every dollar it adds beyond its need is lost on the projects that are not fully funded: its profit is at most $W(I) - (\ell - \sigma_a)$.

Proposition 13. *Consider an MFN clause (ρ_a^0, m) against an unpriced SAFE, which is a fixed share $\rho_a = \rho_a^0$ when $m = 0$, and no protection when $\rho_a^0 = m = 0$. With a contribution σ_a, the accelerator covers $[0, \sigma_a]$ alone, and the seed investor finances exactly the shocks*

$$\sigma_a < \ell \le \sigma_a + (1 - \rho_a^0) W(I) - m.$$

On the last one, the accelerator keeps $\rho_a^0 W(I) + m$. The experiment is covered with probability $G\big(\sigma_a + (1-\rho_a^0)W(I) - m\big)$.

→ Proof in Appendix A, page 39

The seed investor finances a project only if the equity above its amount covers what the accelerator keeps. A protection raises what the accelerator keeps on the last project, and the seed investor stops earlier. A higher contribution moves the whole zone without changing its width, and the probability of a seed round rises or falls depending on whether g is higher at its upper or at its lower end.

Write σ_a^B for the contribution without protection, (σ_a^P, ρ_a^P) for the optimal fixed share, $(\sigma_a^M, \rho_a^{0M}, m^M)$ for an optimal MFN clause, and $x^*(\ell)$ for the threshold the seed investor chooses under the MFN clause.

Proposition 14. *(i) Moving from no protection to the optimal fixed share changes the probability of full funding by*

$$\underbrace{\int_{\sigma_a^B}^{\sigma_a^P} \Big[\int_0^s f(I+\ell-s)\, g(\ell)\, d\ell + \big(1-F(I)\big) g\big(s+W(I)\big) \Big] ds}_{\textit{contribution}} - \underbrace{\big(1-F(I)\big)\Big[G\big(\sigma_a^P + W(I)\big) - G\big(\sigma_a^P + (1-\rho_a^P)W(I)\big)\Big]}_{\textit{last project} \,>\, 0}.$$

The first term has the sign of $\sigma_a^P - \sigma_a^B$: if the fixed share does not raise the contribution, it strictly lowers the probability of full funding.

(ii) With an MFN clause, on a round where the seed investor invests beyond its need, the project is fully funded with probability $1 - F(x^*) > 1 - F(I)$.

→ Proof in Appendix A, page 40

A dollar more from the accelerator raises the probability of full funding twice: its unused cash lowers the threshold on the projects it finances alone, and the seed investor's zone moves to larger shocks without shrinking. The fixed share leaves the accelerator $\rho_a^P W(I)$ on the last project, and the shocks between $\sigma_a^P + (1-\rho_a^P)W(I)$ and $\sigma_a^P + W(I)$ are no longer covered.

Proposition 15. *Moving from the optimal fixed share to an optimal MFN clause changes the probability of a seed round by*

$$\underbrace{\big[G\big(\sigma_a^M + (1-\rho_a^P)W(I)\big) - G(\sigma_a^M)\big] - \big[G\big(\sigma_a^P + (1-\rho_a^P)W(I)\big) - G(\sigma_a^P)\big]}_{\textit{contribution}} + \underbrace{G\big(\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M\big) - G\big(\sigma_a^M + (1-\rho_a^P)W(I)\big)}_{\textit{last project}},$$

and the probability of full funding by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M}\Big[\int_0^s f(I+\ell-s)\,g(\ell)\,d\ell+\big(1-F(I)\big)g\big(s+(1-\rho_a^P)W(I)\big)\Big]ds}_{\textit{contribution}}$$
$$+\underbrace{\big(1-F(I)\big)\Big[G\big(\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M\big)-G\big(\sigma_a^M+(1-\rho_a^P)W(I)\big)\Big]}_{\textit{last project}}$$
$$+\underbrace{\int_{\sigma_a^M}^{\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M}\big(F(I)-F(x^*(\ell))\big)g(\ell)\,d\ell}_{\textit{investment beyond the need}\ \geq\ 0}.$$

→ Proof in Appendix A, page 40

The last-project terms have the sign of $\rho_a^P W(I)-\rho_a^{0M}W(I)-m^M$, the difference in what the accelerator keeps on the last project, the contribution term of full funding the sign of $\sigma_a^M-\sigma_a^P$, and the contribution term of the seed round no sign. Proposition 10 determines none of them; only the investment beyond the need has a sign.

The investment beyond the need acts on the accelerator's choices. Fix a contribution σ_a and an MFN clause (ρ_a^0, m). On a shock ℓ, write $\Pi_s^*(\ell)$ for the seed investor's profit; it depends on ℓ only through $\ell-\sigma_a$, and so does $x^*(\ell)$.

Proposition 16. *(i) The accelerator's profit is*

$$B(\sigma_a)+\int_{\sigma_a}^{\sigma_a+(1-\rho_a^0)W(I)-m}\Big[W(I)-(\ell-\sigma_a)-\Pi_s^*(\ell)-\int_{x^*(\ell)}^I F(x)\,dx\Big]g(\ell)\,d\ell.$$

(ii) Suppose that the optimal fixed share at σ_a finances the same last shock, $\sigma_a+(1-\rho_a^0)W(I)-m$. The accelerator gains from lowering ρ_a^0, which extends the seed investor's zone, if and only if

$$\int_{\sigma_a}^{\sigma_a+(1-\rho_a^0)W(I)-m}\Big[W(I)+\frac{\partial}{\partial\rho_a^0}\Big(\Pi_s^*(\ell)+\int_{x^*(\ell)}^I F(x)\,dx\Big)\Big]g(\ell)\,d\ell>0.$$

On a round where the seed investor invests only its need, the integrand is $mW(I)/(\ell-\sigma_a+m)>0$. On a round where it invests more, the integrand contains $F(x^*)\,\partial\sigma_s^*/\partial\rho_a^0<0$.

(iii) With $\bar\ell=\sigma_a+(1-\rho_a^0)W(I)-m$ the last shock the seed investor finances, the derivative of the profit in σ_a is

$$\big(\rho_a^0W(I)+m\big)g(\bar\ell)-\big(1-G(\bar\ell)\big)-\int_0^{\sigma_a}F(I+\ell-\sigma_a)\,g(\ell)\,d\ell$$
$$-\int_{\sigma_a}^{\bar\ell}\Big[\Pi_s^*(\ell)+\int_{x^*(\ell)}^I F(x)\,dx\Big]g'(\ell)\,d\ell.$$

→ Proof in Appendix A, page 41

By (i), the accelerator keeps, on each shock the seed investor finances, the value of the project at the threshold I net of the round, minus the seed investor's profit, minus the surplus lost by the investment beyond the need, $\int_{x^*}^{I} F(x)\,dx$: each extra dollar lowers the threshold, and is lost on the projects that are not fully funded, with probability $F(x)$ at threshold x.

Point (ii) is the last-project channel. The MFN clause takes more from small rounds than a fixed share with the same last shock: on a round without investment beyond the need, a stronger protection lowers the seed investor's profit by less than $W(I)$, and the accelerator gains from extending the zone. The investment beyond the need pulls the other way: a stronger protection curbs it, which recovers lost surplus, and the accelerator gains from protecting more and covering fewer shocks.

Point (iii) is the contribution channel. On the seed investor's projects, the accelerator's dollar replaces a dollar of the seed investor, and the accelerator recovers it; it is lost on the shocks no one covers, and, when not fully funded, on the projects the accelerator finances alone. Raising σ_a also shifts the seed investor's zone towards larger shocks, and with it the seed investor's profit and the lost surplus, which move to shocks of a different density. With the fixed share, the seed investor's profit, $(1-\rho_a)W(I) - (\ell - \sigma_a)$, falls linearly with the round, and the condition reduces to the one of Proposition 6. The MFN clause lowers the seed investor's profit most on small rounds, and the investment beyond the need adds lost surplus on these same rounds. Whether the accelerator invests more or less than with a fixed share depends on these two profiles and on the slope of g over the zone.

7 Total surplus

The probability of full funding counts every fully funded project the same, whatever its value. The total surplus is the sum of the payoffs of the accelerator, the seed investor and the equity investor, whose profit is zero. The transfers between them cancel: on a covered project with threshold x, the total surplus is the value of the equity minus all the money brought before the equity round, $W(x) - (I + \ell - x)$. On a project abandoned for lack of a seed round, it is $-\sigma_a$, the contribution spent. Since $W' = -(1-F)$, the surplus of a covered project equals

$$W(I) - \ell - \int_{x}^{I} F(t)\,dt.$$

The last term is the cost of the money brought beyond the need, $I - x$. Integrating by parts,

$$\int_{x}^{I} F(t)\,dt = (I-x)\,F(x) + \int_{x}^{I} (I-\pi)\,f(\pi)\,d\pi :$$

this money is lost on the projects that are abandoned, $\pi < x$, and finances projects worth less than the scaling cost, $x < \pi < I$.

The surplus of a covered project is at most $W(I) - \ell$, reached at the threshold I, and the surplus of an abandoned project is at most zero. The first best contributes nothing and covers exactly the shocks $\ell \leq W(I)$, at the threshold I:

$$TS^* = \int_0^{W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell.$$

Proposition 17. *With an MFN clause (ρ_a^0, m), which covers the fixed share and the absence of protection (Proposition 13), let $\bar{\ell} = \sigma_a + (1-\rho_a^0)W(I) - m$ be the last shock the seed investor finances. The total surplus is*

$$TS = TS^* - L_{acc} - L_{cov} - L_{over} - L_{sunk},$$

where

$$L_{acc} = \int_0^{\sigma_a} \int_{I+\ell-\sigma_a}^{I} F(t)\, dt\; g(\ell)\, d\ell, \qquad L_{cov} = \int_{\bar{\ell}}^{W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell \geq 0,$$
$$L_{over} = \int_{\sigma_a}^{\bar{\ell}} \int_{x^*(\ell)}^{I} F(t)\, dt\; g(\ell)\, d\ell, \qquad L_{sunk} = \sigma_a\big(1 - G(\bar{\ell})\big).$$

When the seed investor invests only its need, a dollar more of contribution, at a given amount kept on the last project, changes the total surplus by

$$\big(\rho_a^0 W(I) + m\big) g(\bar{\ell}) - \big(1 - G(\bar{\ell})\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell,$$

and a dollar more kept on the last project, at a given contribution, by $-(\rho_a^0 W(I) + m)\, g(\bar{\ell})$.

→ Proof in Appendix A, page 42

L_{acc} is the cost of the money the accelerator brings beyond the shock, L_{over} the cost of the money the seed investor brings beyond its need, and L_{sunk} the contribution spent on the abandoned projects. L_{cov} is a loss in both directions: if $\bar{\ell} < W(I)$, the shocks with positive NPV that the seed investor no longer covers; if $\bar{\ell} > W(I)$, the shocks with negative NPV that it covers because the accelerator's contribution is already spent.

Without protection, the accelerator keeps nothing on the last project, and the total surplus falls with the contribution. The marginal gain of the accelerator in Proposition 3, $W(I)\, g(\sigma_a)$, is a project it takes from the seed investor, who would have financed it at the threshold I. The accelerator invests whenever $\sigma_a^B > 0$, while the total surplus is maximal at $\sigma_a = 0$. With an uncapped SAFE without discount, the seed investor finances no shock, and the total surplus also loses the shocks in $(\sigma_a^B, \sigma_a^B + W(I)]$.

Proposition 18. *Moving from no protection to the optimal fixed share changes the total surplus by*

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^B}\Big[\big(1-G(s+W(I))\big)+\int_0^s F(I+\ell-s)\,g(\ell)\,d\ell\Big]ds}_{\text{contribution}} - \underbrace{\int_{\sigma_a^P+(1-\rho_a^P)W(I)}^{\sigma_a^P+W(I)}\big(\sigma_a^P+W(I)-\ell\big)\,g(\ell)\,d\ell}_{\text{last project}>0}.$$

The first term has the sign of $\sigma_a^B-\sigma_a^P$.

→ Proof in Appendix A, page 43

Proposition 19. *Moving from the optimal fixed share to an optimal MFN clause changes the total surplus by*

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M}\Big[\rho_a^P W(I)\,g\big(s+(1-\rho_a^P)W(I)\big)-\big(1-G(s+(1-\rho_a^P)W(I))\big)-\int_0^s F(I+\ell-s)\,g(\ell)\,d\ell\Big]ds}_{\text{contribution}} + \underbrace{\int_{\sigma_a^M+(1-\rho_a^P)W(I)}^{\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M}\big(\sigma_a^M+W(I)-\ell\big)\,g(\ell)\,d\ell}_{\text{last project}} - \underbrace{\int_{\sigma_a^M}^{\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M}\int_{x^*(\ell)}^{I}F(t)\,dt\;g(\ell)\,d\ell}_{\text{investment beyond the need}\geq 0}.$$

→ Proof in Appendix A, page 43

The last-project term has the sign of $\rho_a^P W(I)-\rho_a^{0M}W(I)-m^M$, and the contribution term no sign. On each shock the seed investor finances in addition, the total surplus gains $\sigma_a+W(I)-\ell$, the value of the project at the threshold I net of the round, against $-\sigma_a$ if it were abandoned. The investment beyond the need raises the probability of full funding (Proposition 14) and lowers the total surplus: the projects it brings above the threshold are worth less than I.

8 Software and deep tech

Software and deep tech. We sort projects by the size of their liquidity needs relative to their scaling cost, $E(\ell)/I$. A software-like project has a small ratio: its

need is mostly the scaling cost. A deep-tech project has a ratio above one: its need comes mostly as liquidity shocks.

Setting. The model is homogeneous of degree one in $(\pi, I, \ell, \sigma_a, m, \sigma_s)$, so we normalize $I = 1$. We take a uniform f on $[0, 1.8]$, so that $E(\pi) = 0.9 < I$ and $W(I) \approx 0.178$. For ℓ, we take three log-concave distributions of mean $E(\ell)$, from $0.015\,I$ to $4I$: Gamma with shape 2 and 5, and Weibull with shape 3. The benchmark contribution maximizes B on a grid. For the fixed share, $m = 0$, and for the MFN clause, each imposed m/σ_a on a grid, we maximize over σ_a and ρ_a^0, the seed investor's answer being computed round by round with an unpriced SAFE. We also let the fixed part of the clause be the accelerator's own SAFE, on the rest of the contribution, $\sigma_a - m$.

Contracts. The configurations split into two regimes (Figures 1 and 2). For software-like projects, $E(\ell)/I \leq 0.05$, the accelerator invests: 0.03 to $0.10\,I$ without protection, less with the fixed share, and less again with the MFN clause. Almost all of its money goes through the clause, $m^*/\sigma_a^* = 1$, and 0.95 for the Weibull at $E(\ell)/I = 0.05$. The MFN clause gives the highest profit. When the accelerator invests, the optimal fixed share keeps 0.125 to 0.168 per seed project, well above $W(I) - W(I + \sigma_a^P) \leq 0.028$, what a capped SAFE of the same contribution keeps at least: the SAFE would do as well. With the accelerator's own SAFE as its fixed part, the optimal clause earns at most $2 \cdot 10^{-5}$ more, within numerical precision: the value the SAFE keeps in the low states is worth nothing. From $E(\ell)/I = 0.15$ on, the accelerator invests nothing under any protection: few shocks are small enough for it to cover them alone, and on all the others its money goes to the seed investor or is lost. The MFN component is then zero, since $m \leq \sigma_a$, and the protection is a fixed share between 0.17 and 0.45 of a project the accelerator does not fund.

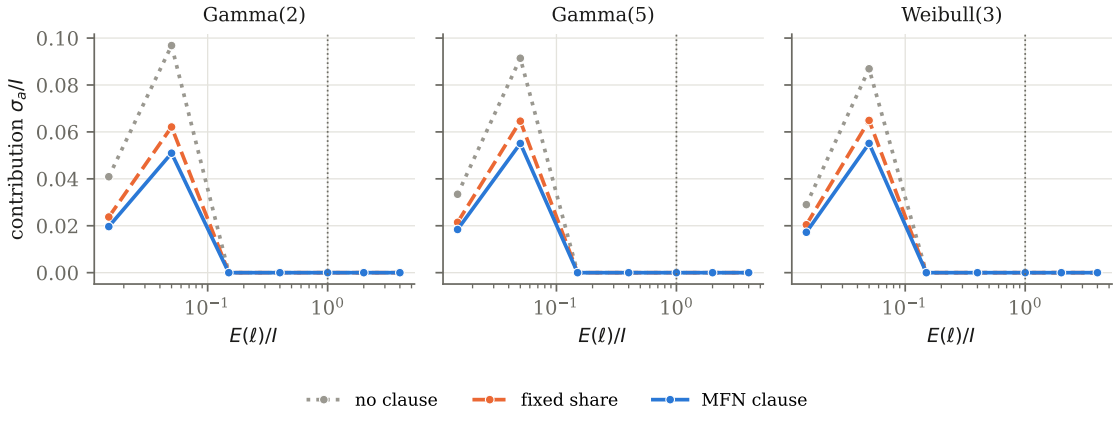


Figure 1: Accelerator's contribution σ_a/I without protection, at the optimal fixed share and at the optimal MFN clause.

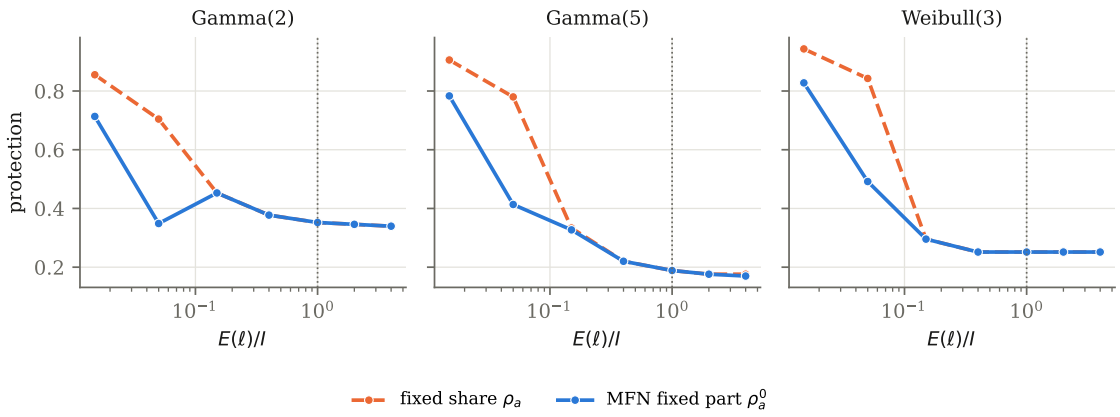


Figure 2: Optimal fixed share ρ_a^* and fixed part of the optimal MFN clause ρ_a^{0*}.

Total surplus. For software-like projects, the MFN clause gives a higher total surplus than the fixed share, which gives a higher one than no protection: 0.88–0.98 of the first best with the clause, 0.86–0.98 with the fixed share, 0.79–0.95 without protection (Figure 3). The protections lower the contribution, which the accelerator brings beyond what maximizes total surplus. From $E(\ell)/I = 0.15$ on, the absence of protection reaches the first best: the accelerator invests nothing, and the seed investor, which takes the whole project, covers exactly the shocks below $W(I)$ at the threshold I. The protection removes the shocks between $(1 - \rho_a)W(I)$ and $W(I)$, and the total surplus falls to 0.72–0.77 of the first best.

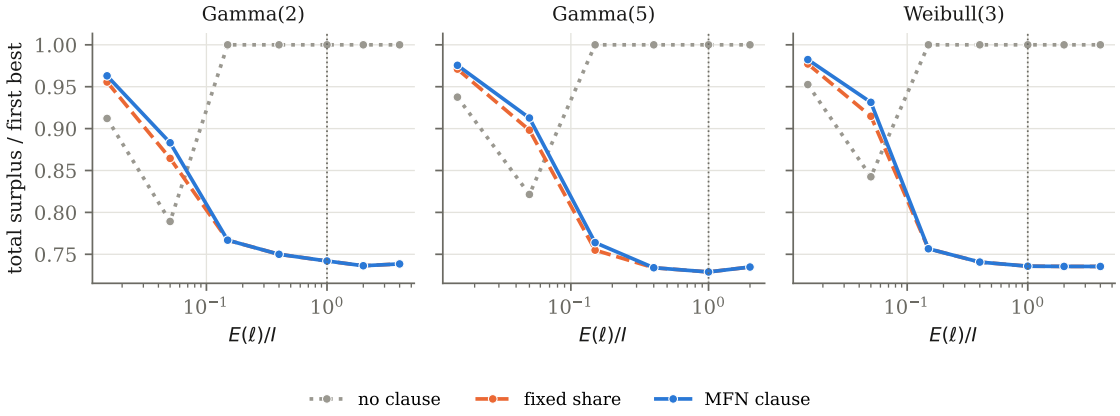


Figure 3: Total surplus relative to the first best, in the configurations where the first best exceeds 10^{-6}.

Seed rounds and full funding. The probabilities show which projects survive. For software-like projects, protection raises the probability of a seed round, from 0.006–0.10 without protection to 0.16–0.23 with the fixed share and 0.26–0.38 with the MFN clause, since the accelerator invests less and leaves more shocks to the seed

investor. It slightly lowers the probability of full funding, from 0.45–0.47 to 0.43–0.45 (Figures 4 and 5). From $E(\ell)/I = 0.15$ on, the accelerator invests nothing, and the protection only shortens the seed investor's zone: it roughly halves both probabilities. At $E(\ell)/I = 0.15$, the probability of a seed round falls from 0.69–0.71 to 0.34–0.37, and that of full funding from 0.30–0.31 to 0.15–0.17. By Proposition 12, both probabilities vanish as the shocks move above $W(I)$.

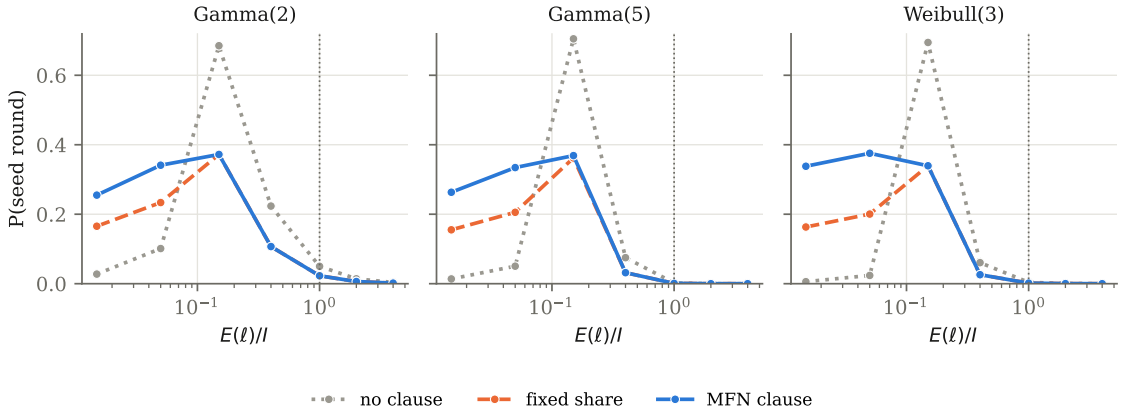


Figure 4: Probability of a seed round.

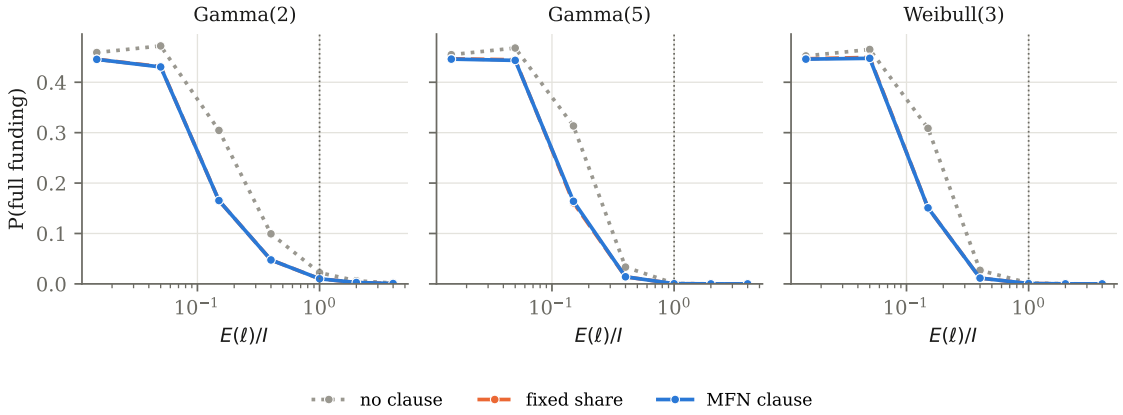


Figure 5: Probability of full funding.

9 Robustness

Another distribution of the project's value. We replace the uniform f by a Beta$(2, 2)$ distribution stretched over $[0, 1.8]$, which is log-concave, has the same mean $E(\pi) = 0.9$, and gives $W(I) \approx 0.123$. The two regimes are unchanged. For software-like projects, the MFN clause gives the highest profit, $m^*/\sigma_a^* = 0.95$ to 1, and its fixed part lies between 0.22 and 0.76; the total surplus is 0.82–0.97 of the first

best with the clause, 0.80–0.97 with the fixed share and 0.71–0.93 without protection. From $E(\ell)/I = 0.15$ on, the accelerator invests nothing, its fixed share lies between 0.18 and 0.42, and it lowers the total surplus to 0.74–0.76 of the first best, which the absence of protection reaches.

The founders' stake. So far the entrepreneur sells the whole project. Suppose instead that the founders keep a share ρ_e of the equity held before the equity round, which they need to keep working on the project. Their stake is written before the seed round, and comes first, with the accelerator's protection: the seed investor's share may not exceed $1 - \rho_e - \rho_a$ with a fixed share, and $1 - \rho_e$ without protection. On the shocks it covers alone, the accelerator keeps $(1 - \rho_e)W(I + \ell - \sigma_a)$.

Proposition 20 (The founders' stake)**.** *With a founders' stake ρ_e, the seed investor's answers of Propositions 1, 4, 5, 8 and 9 hold with ρ_a replaced by $\rho_e + \rho_a$, ρ_a^0 by $\rho_e + \rho_a^0$, and no protection by a fixed share ρ_e. Without protection, the seed investor finances the shock if and only if $\ell - \sigma_a \leq (1 - \rho_e)W(I)$, with the cap $\sigma_s^*/(1 - \rho_e) > \sigma_s^*$, and with $\sigma_a = 0$ the total surplus is*

$$TS^* - \int_{(1-\rho_e)W(I)}^{W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell < TS^*.$$

→ Proof in Appendix A, page 43

The seed investor's cap now exceeds its amount, and the absence of protection no longer reaches the first best: the founders' stake shortens the seed investor's zone, as a fixed share does.

We compute the optimal contracts with $\rho_e = 0.25$ and $\rho_e = 0.5$, the uniform f, and $E(\ell)/I$ from 0.015 to 1. With $\rho_e = 0.25$, the two regimes are unchanged. For software-like projects, the MFN clause gives the highest profit, with $m^* = \sigma_a^*$ or nearly, and a total surplus of 0.86–0.98 of the first best, against 0.85–0.98 with the fixed share and 0.85–0.96 without protection. From $E(\ell)/I = 0.15$ on, the accelerator invests nothing, the absence of protection gives 0.59–0.94 of the first best, and the fixed share roughly halves it, to 0.30–0.62.

With $\rho_e = 0.5$, a third regime appears at $E(\ell)/I = 0.05$ with Gamma shocks. The optimal fixed share comes with no contribution, while the optimal MFN clause keeps the accelerator investing, $\sigma_a^* = 0.03$ to 0.04, all of it through the clause. The MFN clause gives 0.79–0.87 of the first best, against 0.62–0.67 with the fixed share and 0.90–0.92 without protection. The boundary between the regimes therefore depends on the protection, and not only on $E(\ell)/I$.

10 Discussion

The value of the accelerator's money. In the model, the accelerator already owns the project, and its contribution has value only on the shocks it covers alone: on the others, the seed investor takes it with the project, or it is lost. The contribution does not buy a stake from founders, nor does it pay for the program's services. Either would give the contribution a value on every project, and would let the MFN clause operate where, in our simulations, the accelerator invests nothing. The founders' stake of Section 9 is a first step: when the founders keep half of the equity, the clause keeps the accelerator investing where a fixed share does not.

The seed investor's preferences. At a given protection, the MFN clause lowers the seed investor's profit. At the accelerator's optimal protection, the comparison is open: the clause lowers the seed investor's profit per project, but changes the set of projects it finances.

Competitive seed investors. If seed investors earn zero profit, no protection has value: the seed investor's share already leaves the accelerator all the surplus it can take.

The global optimal protection. We compare fixed shares and MFN clauses, the protections that accelerators use. The optimal protection among all contracts is not characterized: the shock is not contractible, and the seed investor answers through its amount. We leave it, and the equity rounds with pro rata rights, for further research.

11 Conclusion

We model the accelerator's protection against a seed investor with market power, and use the model to evaluate the MFN clause that Y Combinator added to its standard deal in 2022.

We summarize. First, without protection, the seed investor takes all of the equity of the projects it finances, with a priced SAFE, a capped SAFE, or an uncapped SAFE whose discount it chooses; it never uses an uncapped SAFE without discount. The accelerator, whose contribution is spent, keeps value only on the shocks it covers alone. Second, a fixed post-money share gives the accelerator this share of every project the seed investor finances, priced and unpriced SAFEs give the same outcome, and a capped SAFE for the accelerator does no better than the fixed share. Third, when the optimal fixed share comes with a contribution, an MFN component strictly beats any fixed share, against a priced or an unpriced SAFE, because it discriminates

between rounds; against the clause, the seed investor prefers the unpriced SAFE, invests beyond its need in the smallest rounds, and the fixed share survives only to curb this answer. Fourth, a seed round covers at most the project's expected surplus at the scaling cost, which is below the scaling cost. The protections change the probabilities of a seed round and of full funding through the contribution and through what the accelerator keeps on the last project, and the investment beyond the need, which raises the probability of full funding, lowers the total surplus. Fifth, numerically, the accelerator invests and puts almost all of its money through the clause only when liquidity needs are small relative to the scaling cost; for larger needs, it invests nothing, and its fixed share lowers the total surplus to 0.72–0.77 of the first best, which the absence of protection reaches.

The optimal contracts take two forms. For software-like projects, all of the accelerator's money converts through the MFN component, and the fixed part of the clause, between 0.35 and 0.83, is guaranteed on top of it, as a share or as the accelerator's own SAFE. For projects whose liquidity needs are larger, the accelerator brings no money and keeps a fixed share between 0.17 and 0.45, which the seed investor's terms cannot dilute.

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A Proofs

Throughout, $x = I + \ell - \sigma_a - \sigma_s$, and since $\partial x/\partial \sigma_s = -1$, the equity value $W(x)$ has derivative $1 - F(x)$ in σ_s. By Leibniz, $W'(x) = -(1 - F(x))$: the boundary term vanishes because the integrand is zero at $\pi = x$.

Proof of Proposition 1

The seed investor solves

$$\max_{\sigma_s,\, \rho_s} \Pi_s = \rho_s\, W(x) - \sigma_s$$

subject to (L) $\sigma_s \geq \ell - \sigma_a$ and $\rho_s \in [0, 1]$. The accelerator accepts any offer, since rejecting it gives it nothing.

The share. Π_s strictly increases in ρ_s as soon as $W(x) > 0$: the seed investor takes $\rho_s = 1$, and $\Pi_s = W(x) - \sigma_s$.

(L) binds. This profit has derivative $(1 - F(x)) - 1 = -F(x) \leq 0$ in σ_s. One more unit of σ_s costs one unit and is returned only in the states where the project is fully funded. The seed investor goes down to $\sigma_s = \ell - \sigma_a$, where $x = I$, and earns $W(I) - (\ell - \sigma_a)$. If this is negative, no larger amount does better. The seed investor therefore invests if and only if $\ell - \sigma_a \leq W(I)$, and the accelerator keeps $(1 - \rho_s^*)W(I) = 0$. □

Proof of Proposition 2

(i) The uncapped SAFE without discount. In state $\pi > x$, the SAFE receives $\min(\sigma_s/(\pi - x), 1)(\pi - x) = \min(\sigma_s, \pi - x)$. We show that the seed investor loses money at every amount. Its gross payoff is $\pi - x$ on $x < \pi < x + \sigma_s$ and σ_s above, with $F = 1$ above 1:

$$\int_x^{x+\sigma_s} (\pi - x)\, f(\pi)\, d\pi + \sigma_s\big(1 - F(x + \sigma_s)\big).$$

Integrating the first term by parts, it equals $\sigma_s F(x + \sigma_s) - \int_x^{x+\sigma_s} F(\pi)\, d\pi$, and the gross payoff is $\sigma_s - \int_0^{\sigma_s} F(x + t)\, dt$. Hence $\Pi_s = -\int_0^{\sigma_s} F(x + t)\, dt$. Since $x \geq 0$ and $f > 0$, $F(x + t) > 0$ for every $t > 0$, and $\Pi_s < 0$. The project is abandoned, and the accelerator keeps nothing.

(ii) The other SAFEs. The seed investor's gross payoff is at most the equity, $W(x)$, so its profit is at most $W(x) - \sigma_s$, which has derivative $-F(x) \leq 0$ in σ_s: it is at most $W(I) - (\ell - \sigma_a)$, the profit of Proposition 1. Take $\sigma_s = \ell - \sigma_a$, so that $x = I$.

Uncapped SAFE with discount. The SAFE receives $\min\big((\ell-\sigma_a)/\lambda, \pi - I\big)$. With $\lambda = \min\big(1, (\ell-\sigma_a)/(1-I)\big)$, the face value $(\ell-\sigma_a)/\lambda$ is at least $1 - I \geq \pi - I$, and the SAFE receives the whole equity in every state.

SAFE with a cap. With a cap $V_s \leq \ell - \sigma_a$ and any discount, the claim $(\ell - \sigma_a)/\min(V_s, \lambda(\pi - I))$ is at least $(\ell - \sigma_a)/V_s \geq 1$ in every state, and the SAFE receives the whole equity.

In both cases, the seed investor attains $W(I) - (\ell - \sigma_a)$, invests if and only if $\ell - \sigma_a \leq W(I)$, and the accelerator keeps nothing. □

Proof of Proposition 3

An optimum exists. B is continuous and $B(0) = 0$. For $\sigma_a \geq I$, $W(I + \ell - \sigma_a) \leq W(I - \sigma_a) = E(\pi) - I + \sigma_a$, so $B(\sigma_a) \leq G(\sigma_a)(E(\pi) - I + \sigma_a) - \sigma_a < 0$ by (A1). The maximum is reached on $[0, I)$.

The first-order condition. By Leibniz, the boundary term evaluates the integrand at $\ell = \sigma_a$, where the threshold is I, and the inner term uses $\partial W(I + \ell - \sigma_a)/\partial \sigma_a = 1 - F(I + \ell - \sigma_a)$:

$$\begin{aligned} B'(\sigma_a) &= W(I)\, g(\sigma_a) + \int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell - 1 \\ &= W(I)\, g(\sigma_a) - \big(1 - G(\sigma_a)\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell. \end{aligned}$$

At a positive optimum, $B'(\sigma_a^B) = 0$, which is the stated condition. □

Proof of Proposition 4

The accelerator holds a post-money share ρ_a, which the seed round does not dilute. The seed investor solves

$$\max_{\sigma_s,\, \rho_s} \Pi_s = \rho_s\, W(x) - \sigma_s$$

subject to (L) $\sigma_s \geq \ell - \sigma_a$ and (PM) $\rho_s \leq 1 - \rho_a$.

The share. Π_s strictly increases in ρ_s as soon as $W(x) > 0$: (PM) binds, $\rho_s = 1 - \rho_a$.

The amount. $(1 - \rho_a)W(x) - \sigma_s$ has derivative $(1 - \rho_a)(1 - F(x)) - 1 < 0$ in σ_s. Hence $\sigma_s^* = \ell - \sigma_a$ and $x^* = I$. The seed investor earns $(1 - \rho_a)W(I) - (\ell - \sigma_a)$, invests if and only if this is nonnegative, and the accelerator keeps $\rho_a W(I)$. □

Proof of Proposition 5

By the contracts of Section 2, the seed investor receives, in state π, the share

$$\min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ 1 - \rho_a\Big)$$

of the equity $\pi - x$: its claim, capped by the fixed share.

Analysis. Its share never exceeds $1 - \rho_a$, so its profit is at most $(1 - \rho_a)W(x) - \sigma_s$, which has derivative $(1 - \rho_a)(1 - F(x)) - 1 < 0$ in σ_s. The bound is largest at $\sigma_s = \ell - \sigma_a$, where $x = I$: $\Pi_s \leq (1 - \rho_a)W(I) - (\ell - \sigma_a)$, the profit of Proposition 4.

Synthesis. Take $\sigma_s = \ell - \sigma_a$ and a cap $V_s \leq (\ell - \sigma_a)/(1 - \rho_a)$. With any discount,

$$\frac{\ell - \sigma_a}{\min(V_s, \lambda(\pi - I))} \geq \frac{\ell - \sigma_a}{V_s} \geq 1 - \rho_a,$$

so the share equals $1 - \rho_a$ in every state $\pi > I$, and the seed investor attains the bound. A discount brings nothing. □

Proof of Proposition 6

By Propositions 4 and 5, the program is the same against a priced and an unpriced SAFE.

(i) The share. At given σ_a, the term $\rho_a W(I)[G(\sigma_a + (1-\rho_a)W(I)) - G(\sigma_a)]$ vanishes at $\rho_a = 0$ and at $\rho_a = 1$, where the seed investor finances nothing, and is strictly positive in between, since $g > 0$. Hence ρ_a^* is interior and the derivative in ρ_a vanishes; dividing by $W(I)$ gives the stated condition. On the left, what a stronger protection returns on each project the seed investor finances; on the right, what the accelerator kept on the marginal project, which the seed investor stops financing.

We show that this term is strictly log-concave in ρ_a, so that the condition has a single solution. Its first factor, ρ_a, is log-concave. Its second factor is $G(\sigma_a + (1-\rho_a)W(I)) - G(\sigma_a)$, the difference between G at a point affine in ρ_a and $G(\sigma_a)$. At a point $\ell > \sigma_a$, its log-derivative is

$$\frac{g(\ell)}{G(\ell) - G(\sigma_a)} = \Big[\int_0^{\ell - \sigma_a} \frac{g(\ell - s)}{g(\ell)}\, ds\Big]^{-1}.$$

When ℓ rises, the bound rises and each ratio $g(\ell - s)/g(\ell)$ rises, its log-derivative being positive since $(\log g)'$ decreases (Bagnoli and Bergstrom, 2005). The integral rises, and the log-derivative falls: the second factor is log-concave, and the product is strictly log-concave since the first factor is. The condition characterizes the unique maximum $\rho_a^*(\sigma_a)$.

(ii) The contribution. An optimum exists: Π_a is continuous, $\Pi_a(0,\rho_a) \geq 0$, and for $\sigma_a \geq I$ the bound of the proof of Proposition 3 gives $\Pi_a \leq (1-G(\sigma_a))(W(I)-\sigma_a) - G(\sigma_a)(I-E(\pi)) < 0$, since $W(I) < I$. The maximum is reached on the compact set $[0,I]\times[0,1]$. At fixed ρ_a, the derivative of Π_a in σ_a is

$$\begin{aligned} B'(\sigma_a) + \rho_a W(I)\Big[g\big(\sigma_a + (1-\rho_a)W(I)\big) - g(\sigma_a)\Big] \\ = (1-\rho_a)W(I)\,g(\sigma_a) + \rho_a W(I)\,g\big(\sigma_a + (1-\rho_a)W(I)\big) \\ - \big(1-G(\sigma_a)\big) - \int_0^{\sigma_a} F(I+\ell-\sigma_a)\,g(\ell)\,d\ell, \end{aligned}$$

by the expression of B' in the proof of Proposition 3. At a positive optimum, it vanishes at $\rho_a = \rho_a^*$, by the envelope theorem.

(iii) The comparison with the benchmark. At σ_a^B, any $\rho_a \in (0,1)$ adds $\rho_a W(I)[G(\sigma_a^B + (1-\rho_a)W(I)) - G(\sigma_a^B)] > 0$ to $B(\sigma_a^B)$. Hence $\Pi_a(\sigma_a^*,\rho_a^*) > B(\sigma_a^B)$. □

Proof of Proposition 7

What the protection leaves. We show that, in the state $\pi > x$, the protection leaves

$$\Big(\pi - x - \frac{\sigma_a}{\lambda_a}\Big)^+ - \frac{\sigma_a}{V_a}\Big(\pi - x - \frac{V_a}{\lambda_a}\Big)^+.$$

Since $1/\min(a,b) = \max(1/a, 1/b)$, the claim times $\pi - x$ is $\max(\sigma_a(\pi-x)/V_a,\ \sigma_a/\lambda_a)$, and the protection takes this value, at most $\pi - x$. Since $V_a \geq \sigma_a$, $\sigma_a/\lambda_a \leq V_a/\lambda_a$, and there are three zones. If $\pi - x \leq \sigma_a/\lambda_a$, the protection takes everything. If $\sigma_a/\lambda_a \leq \pi - x \leq V_a/\lambda_a$, then $\sigma_a(\pi-x)/V_a \leq \sigma_a/\lambda_a$, and the protection leaves $\pi - x - \sigma_a/\lambda_a$. If $\pi - x \geq V_a/\lambda_a$, it leaves $(1-\sigma_a/V_a)(\pi-x)$. The stated expression takes these three values: in the third zone, it equals $\pi - x - \sigma_a/\lambda_a - (\sigma_a/V_a)(\pi-x) + \sigma_a/\lambda_a$. Since $\int_x^1 (\pi - x - y)^+ f(\pi)\,d\pi = W(x+y)$ for $y \geq 0$, by the definition of W, the protection leaves $W(x+\sigma_a/\lambda_a) - (\sigma_a/V_a)W(x+V_a/\lambda_a)$ in expectation.

The seed investor. *Analysis.* Its share is at most what the protection leaves, so its profit is at most $W(x+\sigma_a/\lambda_a) - (\sigma_a/V_a)W(x+V_a/\lambda_a) - \sigma_s$. Since $W' = -(1-F)$ and x falls by one dollar per dollar invested, this bound has derivative

$$-F\Big(x + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a}\Big(1 - F\Big(x + \frac{V_a}{\lambda_a}\Big)\Big) \leq 0$$

in σ_s, and is maximal at the need, $\sigma_s = \ell - \sigma_a$ and $x = I$.

Synthesis. A priced SAFE with $\rho_s = 1$ receives all the equity the protection leaves. An unpriced SAFE with cap $\ell - \sigma_a$ claims $(\ell-\sigma_a)/\min(\ell-\sigma_a, \lambda(\pi - I)) \geq 1$ in every state, and receives it too. Both attain the bound at the need, which is nonnegative if and only if the stated condition holds.

The accelerator. On each project the seed investor finances, the accelerator keeps the value the protection takes at the threshold I,

$$W(I) - W\Big(I + \frac{\sigma_a}{\lambda_a}\Big) + \frac{\sigma_a}{V_a} W\Big(I + \frac{V_a}{\lambda_a}\Big),$$

and the seed investor finances the shocks up to $\sigma_a + W(I)$ minus this value. Hence the accelerator's profit is the profit of Section 4 with $\rho_a W(I)$ replaced by this value: the two protections differ only by the values they allow at a given contribution.

The values the SAFE allows. If $\sigma_a = 0$, the claim is zero, and the accelerator keeps nothing. Suppose $\sigma_a > 0$. We show that the value kept covers $(W(I) - W(I + \sigma_a), W(I)]$.

- The protection leaves at most $W(I + \sigma_a/\lambda_a) \leq W(I + \sigma_a)$, since W decreases and $\lambda_a \leq 1$: the value kept is at least $W(I) - W(I + \sigma_a)$. With $\lambda_a = 1$, it equals $W(I) - W(I + \sigma_a) + (\sigma_a/V_a)W(I + V_a)$, which tends to this bound as $V_a \to +\infty$.
- With $V_a = \sigma_a$, the value kept is $W(I)$.
- The value kept is continuous in (V_a, λ_a) on a connected set: it takes every value between these two bounds.

The comparison with the fixed share. The fixed share allows the values $[0, W(I)]$, which contain those of the SAFE: no SAFE earns more. Let (σ_a^P, ρ_a^P) be an optimal fixed share. If $\sigma_a^P > 0$ and $\rho_a^P W(I) > W(I) - W(I + \sigma_a^P)$, a SAFE with contribution σ_a^P keeps $\rho_a^P W(I)$, and earns the same profit. If $\sigma_a^P = 0$, then $\rho_a^P > 0$ by Proposition 6, and for $\sigma_a > 0$ small enough, $W(I) - W(I + \sigma_a) < \rho_a^P W(I)$: a SAFE with contribution σ_a keeps $\rho_a^P W(I)$, and its profit tends to the profit of the fixed share as $\sigma_a \to 0$, since B and G are continuous. □

Proof of Proposition 8

ρ_a^0 and m are chosen by the accelerator before the round: here, they are parameters.

The rule. Each SAFE claims a share of the equity held before the equity round. If the claims sum to at most $1 - \rho_a^0$, each receives its claim. Otherwise, each claim is multiplied by $1 - \rho_a^0$ divided by the sum of the claims, so that the shares received sum to $1 - \rho_a^0$. In both cases, each claim is multiplied by the minimum of 1 and of $1 - \rho_a^0$ divided by the sum of the claims. The seed investor's share is therefore the minimum of its claim and of its claim times $1 - \rho_a^0$ divided by the sum of the claims.

The shares. *Unpriced SAFE.* The two SAFEs convert at the same price, $\min(V_s, \lambda(\pi - x))$. The seed investor claims $\sigma_s / \min(V_s, \lambda(\pi - x))$, and the claims sum to $(\sigma_s + m)/\min(V_s, \lambda(\pi - x))$. The seed investor's claim times $1 - \rho_a^0$ divided by this sum is

$$\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))} \cdot \frac{(1 - \rho_a^0)\min(V_s, \lambda(\pi - x))}{\sigma_s + m} = \frac{(1 - \rho_a^0)\,\sigma_s}{\sigma_s + m}.$$

By the rule, its share is

$$\min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s + m}\Big).$$

Priced SAFE. The seed investor converts at its valuation σ_s/ρ_s and claims ρ_s. The MFN component converts at $\min(\sigma_s/\rho_s, \pi - x)$ and claims $m/\min(\sigma_s/\rho_s, \pi - x)$.

In the states $\pi - x \geq \sigma_s/\rho_s$, the MFN component claims $m\rho_s/\sigma_s$, and the claims sum to $\rho_s(\sigma_s+m)/\sigma_s$. The claim ρ_s times $1-\rho_a^0$ divided by this sum is $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$, and, by the rule, the seed investor's share is $\min\big(\rho_s, (1-\rho_a^0)\sigma_s/(\sigma_s+m)\big)$.

In the states $\pi - x < \sigma_s/\rho_s$, the MFN component claims $m/(\pi - x)$, and the claims sum to $\rho_s + m/(\pi - x)$. The claim ρ_s times $1-\rho_a^0$ divided by this sum is, multiplying the numerator and the denominator by $\pi - x$,

$$\frac{(1-\rho_a^0)\,\rho_s}{\rho_s + m/(\pi - x)} = \frac{(1-\rho_a^0)\,\rho_s(\pi - x)}{\rho_s(\pi - x) + m},$$

and, by the rule, the seed investor's share is

$$\min\Big(\rho_s,\ \frac{(1-\rho_a^0)\,\rho_s(\pi - x)}{\rho_s(\pi - x) + m}\Big).$$

We show that, when $m > 0$, this share is strictly below $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. The function $t \mapsto t/(t+m)$ has derivative $m/(t+m)^2 > 0$, so it is strictly increasing. In these states, $\rho_s(\pi - x) < \sigma_s$. Hence $\rho_s(\pi - x)/(\rho_s(\pi - x) + m) < \sigma_s/(\sigma_s + m)$, and, multiplying by $1-\rho_a^0$, the second term of the minimum is strictly below $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. The share, at most this second term, is too.

The cap on the share. In each of the three cases, the seed investor's share is a minimum whose second term is at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$.

Analysis. *The bound.* Under both contracts, the share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. What the seed investor receives, its share times $\pi - x$ in the states $\pi > x$, therefore has an expectation of at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ times $W(x)$, and the profit at the amount σ_s is at most

$$\bar{\Pi}_s(\sigma_s) = \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s + m}\,W(x) - \sigma_s.$$

Its derivative in σ_s. The cap $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ has derivative $(1-\rho_a^0)m/(\sigma_s+m)^2$. Since $x = I + \ell - \sigma_a - \sigma_s$ falls by one dollar per dollar invested and $W'(x) = -(1-F(x))$, $W(x)$ has derivative $1 - F(x)$ in σ_s. The derivative of the bound is therefore

$$(1-\rho_a^0)\Big[\frac{m}{(\sigma_s+m)^2}W(x) + \frac{\sigma_s\big(1-F(x)\big)}{\sigma_s+m}\Big] - 1.$$

The first term is the gain in share: each dollar the seed investor adds dilutes the MFN component, and this gain is large when m weighs a lot against σ_s.

The bound does not rise beyond $x = 0$. We show that its derivative is nonpositive when $x \le 0$, that is, when $\sigma_s \ge I + \ell - \sigma_a$. For $x \le 0$, $F(x) = 0$, and $W(x) = E(\pi) - x = E(\pi) - I - (\ell - \sigma_a) + \sigma_s < \sigma_s$ by (A1). The derivative is therefore at most

$$(1-\rho_a^0)\Big[\frac{m\,\sigma_s}{(\sigma_s+m)^2} + \frac{\sigma_s}{\sigma_s+m}\Big] - 1 = (1-\rho_a^0)\,\frac{\sigma_s(\sigma_s+2m)}{(\sigma_s+m)^2} - 1,$$

which is nonpositive, since $\sigma_s(\sigma_s+2m) = (\sigma_s+m)^2 - m^2$. The bound reaches its maximum on the compact interval $\ell - \sigma_a \le \sigma_s \le I + \ell - \sigma_a$.

The shocks it does not finance. Suppose $\ell - \sigma_a + m > (1-\rho_a^0)W(I)$. Since $W' = -(1-F)$ and $0 \le 1 - F \le 1$,

$$W(x) - W(I) = \int_x^I \big(1 - F(t)\big)\,dt \le I - x = \sigma_s - (\ell - \sigma_a).$$

Multiplying by $1-\rho_a^0$, then using the assumption, $1-\rho_a^0 \le 1$ and $\sigma_s - (\ell - \sigma_a) \ge 0$,

$$\begin{aligned}(1-\rho_a^0)W(x) &\le (1-\rho_a^0)W(I) + (1-\rho_a^0)\big(\sigma_s - (\ell-\sigma_a)\big)\\ &< (\ell-\sigma_a) + m + \sigma_s - (\ell-\sigma_a) = \sigma_s + m.\end{aligned}$$

Multiplying by $\sigma_s/(\sigma_s+m)$, $(1-\rho_a^0)\sigma_s W(x)/(\sigma_s+m) < \sigma_s$: the bound is negative at every $\sigma_s \ge \ell - \sigma_a$, and the seed investor loses money at every amount, with either contract. If $\ell - \sigma_a + m \le (1-\rho_a^0)W(I)$, the bound at the need, $(\ell-\sigma_a)[(1-\rho_a^0)W(I)/(\ell-\sigma_a+m) - 1]$, is nonnegative.

Overinvestment. If the derivative of the bound is positive at $\sigma_s = \ell - \sigma_a$, where $x = I$, the bound rises beyond the need, and its maximum exceeds $\ell - \sigma_a$. This derivative is the left side of the condition of the proposition.

Synthesis. *The unpriced SAFE attains the bound.* Fix σ_s, take $\lambda = 1$ and $V_s = (\sigma_s+m)/(1-\rho_a^0)$. In every state, $\min(V_s, \pi - x) \le V_s$, so $\sigma_s/\min(V_s, \pi - x) \ge \sigma_s/V_s = (1-\rho_a^0)\sigma_s/(\sigma_s+m)$: the minimum giving the share equals its second term, and the share equals $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ everywhere. By the shares above, no (V_s, λ) gives a share above this cap: a discount brings the seed investor nothing. With an unpriced SAFE, the seed investor's profit at the amount σ_s is therefore $\bar{\Pi}_s(\sigma_s)$, and it finances the shock if and only if $\ell - \sigma_a + m \le (1-\rho_a^0)W(I)$.

The priced SAFE does strictly worse when $m > 0$. With $m = 0$, the clause is a fixed share, and both contracts give the same profit (Propositions 4 and 5). Suppose $m > 0$. The priced SAFE's profit is continuous in (σ_s, ρ_s) on a compact set, and has a maximum, at most the maximum of $\bar{\Pi}_s$. Suppose that they are equal, at the priced SAFE's optimum σ_s^p. The states $x < \pi < x + \sigma_s^p/\rho_s$ have positive probability, and in them the priced SAFE's share is strictly below $(1-\rho_a^0)\sigma_s^p/(\sigma_s^p+m)$: its profit is strictly below $\bar{\Pi}_s(\sigma_s^p)$, which contradicts the equality. □

Proof of Proposition 9

The shocks on which the condition holds. We show that the overinvestment condition holds exactly when $\ell - \sigma_a$ is below an explicit bound. Multiplying the condition by $(\ell - \sigma_a + m)^2$, it reads

$$\big[1 - (1-\rho_a^0)(1-F(I))\big](\ell - \sigma_a + m)^2 + (1-\rho_a^0)(1-F(I))\, m\,(\ell - \sigma_a + m) \\ - (1-\rho_a^0)\, m\, W(I) < 0.$$

This is a quadratic in $\ell - \sigma_a + m$, with positive leading coefficient and negative at 0: it is negative exactly below its positive root. This bound depends on f only through $F(I)$ and $W(I)$.

The level of overinvestment. On a shock below this bound, the derivative of the bound vanishes:

$$(1-\rho_a^0)\Big[\frac{m}{(\sigma_s+m)^2}W(x) + \frac{\sigma_s\big(1-F(x)\big)}{\sigma_s+m}\Big] = 1.$$

At a given shock, $x = I + \ell - \sigma_a - \sigma_s$ depends on σ_s, and the equation is implicit. At a given threshold x, multiplying by $(\sigma_s+m)^2$, it becomes a quadratic in σ_s:

$$\big[1-(1-\rho_a^0)(1-F(x))\big]\sigma_s^2 + m\big[2-(1-\rho_a^0)(1-F(x))\big]\sigma_s + m^2 - (1-\rho_a^0)\, m\, W(x) = 0.$$

The constant term is negative as soon as the seed investor's zone is not empty, since $m < (1-\rho_a^0)W(I) \le (1-\rho_a^0)W(x)$ for $x \le I$: there is a single positive root, the one stated in the proposition. The corresponding shock is $\ell - \sigma_a = \sigma_s - (I - x)$, and the accelerator keeps $(1 - (1-\rho_a^0)\sigma_s/(\sigma_s+m))W(x)$. At $x = I$, this root is the bound of the shocks on which the overinvestment condition holds. The threshold x therefore describes these shocks, the amount, the shock and the value kept being explicit functions of x.

On a given shock, the seed investor's critical points are the thresholds x at which this root, added to x, equals $I + \ell - \sigma_a$. If this sum is strictly monotone in x, there is at most one; since the derivative is positive at $\sigma_s = \ell - \sigma_a$, it is the maximum. □

Proof of Proposition 10

By Proposition 8, the seed investor uses an unpriced SAFE.

The program. By Proposition 8, the seed investor finances exactly the shocks $\sigma_a < \ell \le \sigma_a + (1-\rho_a^0)W(I) - m$, where it may invest $\sigma_s^* > \ell - \sigma_a$ and takes $\rho_s^* = (1-\rho_a^0)\sigma_s^*/(\sigma_s^*+m)$. On these projects, the accelerator keeps $(1-\rho_s^*)W(x^*)$, where ρ_s^* and $x^* = I + \ell - \sigma_a - \sigma_s^*$ depend only on $\ell - \sigma_a$, ρ_a^0 and m. Hence

$$\Pi_a(\sigma_a, \rho_a^0, m) = B(\sigma_a) + \int_{\sigma_a}^{\sigma_a + (1-\rho_a^0)W(I) - m} (1-\rho_s^*)W(x^*)\, g(\ell)\, d\ell.$$

For $m = 0$, this is the program of Proposition 6.

The last financed project. There, $\ell - \sigma_a + m = (1 - \rho_a^0)W(I)$. We show that the seed investor invests exactly its need. The inequality used for the shocks it does not finance, with this equality instead of the strict assumption, gives

$$(1 - \rho_a^0)W(x) \le (\ell - \sigma_a) + m + (1 - \rho_a^0)\big(\sigma_s - (\ell - \sigma_a)\big) \le \sigma_s + m,$$

so the bound is nonpositive at every amount and zero at the need. Beyond the need, $x < I$, and the first inequality is strict: $F(I) > 0$ by (A1), F is continuous, hence positive on an interval to the left of I, and $W(x) - W(I) = (I - x) - \int_x^I F(t)\,dt < I - x$. The bound is strictly negative there: the seed investor invests its need and earns zero, and the accelerator keeps $W(I) - (\ell - \sigma_a) = \rho_a^0 W(I) + m$. The last financed project moves back by $W(I)$ per unit of ρ_a^0, and by one unit per unit of m. Lowering ρ_a^0 by one unit and raising m by $W(I)$ leaves it in place.

The MFN component lowers the value kept on no financed project. The seed investor invests at least $\ell - \sigma_a$, so $x^* \le I$ and $W(x^*) \ge W(I)$. Its share is at most $(1 - \rho_a^0)\sigma_s^*/(\sigma_s^* + m) \le 1 - \rho_a^0$. Hence the accelerator keeps at least $\rho_a^0 W(I)$, what it keeps with $m = 0$.

(i) The MFN component is strictly positive. Let (σ_a^P, ρ_a^P) be the optimum of Proposition 6. We show that a small MFN component strictly raises the profit, which rules out $m^* = 0$. Fix a small $\delta > 0$.

When $\sigma_a^P > 0$. Keep $\sigma_a = \sigma_a^P$ and $\rho_a^0 = \rho_a^P$, and add $m \in (0, \sigma_a)$.

- The projects with $\ell - \sigma_a \ge \delta$ and $\ell - \sigma_a + m \le (1 - \rho_a^0)W(I)$ are financed. We show that, for small m, the seed investor invests exactly its need on these projects, using $\rho_a^0 = \rho_a^P > 0$.

  Large excess amounts lose. At $\sigma_s \ge \ell - \sigma_a$, the bound is at most $(1 - \rho_a^0)W(x) - \sigma_s$, since $\sigma_s/(\sigma_s + m) \le 1$. Since $W(x) - W(I) = \int_x^I (1 - F(t))\,dt$ and $I - x = \sigma_s - (\ell - \sigma_a)$,

  $$(1 - \rho_a^0)W(x) - \sigma_s = (1 - \rho_a^0)W(I) - (\ell - \sigma_a) - \int_x^I \big[1 - (1 - \rho_a^0)(1 - F(t))\big]dt,$$

  and the integrand is at least ρ_a^0. At the need, the bound equals $(1 - \rho_a^0)W(I) - (\ell - \sigma_a) - (1 - \rho_a^0)mW(I)/(\ell - \sigma_a + m)$. The bound at σ_s is therefore below the bound at the need as soon as $\sigma_s - (\ell - \sigma_a) > (1 - \rho_a^0)mW(I)/(\rho_a^0\delta)$, a quantity that tends to zero with m.

  Small excess amounts lose too. When $\sigma_s - (\ell - \sigma_a)$ is at most this quantity, x lies within it below I, so $x \ge 0$ for small m and $W(x) \le W(0)$. The derivative of the bound is then at most $(1 - \rho_a^0)[mW(0)/\delta^2 + 1 - F(x)] - 1$, which tends to $(1 - \rho_a^0)(1 - F(I)) - 1 < 0$ as $m \to 0$, uniformly in $\ell - \sigma_a \ge \delta$, since F is continuous. The bound decreases on this interval.

Hence $\sigma_s^* = \ell - \sigma_a$ for small m, and the value kept gains

$$\Big(1 - \frac{(1-\rho_a^0)(\ell-\sigma_a)}{\ell - \sigma_a + m}\Big) W(I) - \rho_a^0 W(I) = \frac{(1-\rho_a^0)\, m}{\ell - \sigma_a + m} W(I).$$

- On the projects with $\ell - \sigma_a < \delta$, the value kept does not fall.
- The projects financed for $m = 0$ and no longer for $m > 0$ lie in a band of width m, and the value kept on each is at most $W(I)$.

The gain of the first group, divided by m, tends to

$$(1-\rho_a^0) W(I) \int_{\sigma_a+\delta}^{\sigma_a + (1-\rho_a^0)W(I)} \frac{g(\ell)}{\ell - \sigma_a}\, d\ell$$

as $m \to 0$, and this quantity tends to $+\infty$ as $\delta \to 0$, since $g(\sigma_a) > 0$. The loss of the third group, divided by m, remains bounded. For δ and then m small enough, the profit strictly rises.

When $\sigma_a^P = 0$ and $g(0) > 0$. The MFN component requires a contribution: take $\sigma_a = m$, $\rho_a^0 = \rho_a^P$. Moving the contribution from 0 to m at $m = 0$ changes the fixed-share profit by at most a constant times m, its derivative in σ_a being bounded. The MFN component then adds, on the rounds $\ell - \sigma_a \geq \delta$, a gain whose ratio to m tends to $(1-\rho_a^0) W(I) \int_{m+\delta}^{m+(1-\rho_a^0)W(I)} g(\ell)/(\ell - m)\, d\ell$, which tends to $+\infty$ as $\delta \to 0$ uniformly in small m, since g is positive and continuous at 0. The losses of the other groups, divided by m, remain bounded, and the profit strictly rises.

In both cases, the argument does not assume that an optimum exists: starting from the optimum with a fixed share, a small MFN component strictly raises the profit, so the accelerator earns strictly more with an MFN clause than with any fixed share, hence than in the benchmark by Proposition 6.

How the value kept reacts to ρ_a^0 and m. On a financed project, the accelerator keeps the project's value minus the seed investor's amount and profit:

$$(1-\rho_s^*) W(x^*) = W(x^*) - \sigma_s^* - \Pi_s(\sigma_s^*).$$

The first two terms have derivative $-F(x^*)$ in σ_s. The value kept therefore has derivative in ρ_a^0

$$-\frac{\partial \Pi_s}{\partial \rho_a^0}(\sigma_s^*) - \Big[F(x^*) + \frac{\partial \Pi_s}{\partial \sigma_s}(\sigma_s^*)\Big] \frac{\partial \sigma_s^*}{\partial \rho_a^0},$$

and the same expression in m. Two kinds of shocks arise.

- Without overinvestment, $\sigma_s^* = \ell - \sigma_a$ does not move, and only the first term remains.

- With interior overinvestment, $\partial\Pi_s/\partial\sigma_s = 0$. Where $\partial^2\Pi_s/\partial\sigma_s^2$ does not vanish, the implicit function theorem gives, at $\sigma_s = \sigma_s^*$ and $x = x^*$,

$$\frac{\partial\sigma_s^*}{\partial\rho_a^0} = \frac{1}{(1-\rho_a^0)\,\partial^2\Pi_s/\partial\sigma_s^2} < 0,$$
$$\frac{\partial\sigma_s^*}{\partial m} = -\frac{(1-\rho_a^0)\big[(\sigma_s - m)W(x) - \sigma_s(\sigma_s+m)(1-F(x))\big]}{(\sigma_s+m)^3\,\partial^2\Pi_s/\partial\sigma_s^2},$$

  since $\partial^2\Pi_s/\partial\sigma_s\partial\rho_a^0 = -(\partial\Pi_s/\partial\sigma_s + 1)/(1-\rho_a^0) = -1/(1-\rho_a^0)$ at the critical point.

With

$$\frac{\partial\Pi_s}{\partial\rho_a^0} = -\frac{\sigma_s W(x)}{\sigma_s + m}, \qquad \frac{\partial\Pi_s}{\partial m} = -\frac{(1-\rho_a^0)\,\sigma_s W(x)}{(\sigma_s+m)^2},$$

a stronger protection always raises the value kept: $\partial\Pi_s/\partial\rho_a^0 < 0$, $\partial\sigma_s^*/\partial\rho_a^0 \le 0$, and $F(x^*) + \partial\Pi_s/\partial\sigma_s \ge 0$. It also curbs overinvestment.

(ii) The smallest rounds overinvest. As $\ell - \sigma_a \to 0$, the left side of the overinvestment condition tends to $(1-\rho_a^0)W(I)/m$, which exceeds 1 since the zone is not empty: $m < (1-\rho_a^0)W(I)$. With $m^* > 0$, there are therefore always shocks on which the seed investor invests beyond its need.

(iii) The split between the two components. With $\rho_a^{0*} > 0$ and $m^* < \sigma_a^*$, both derivatives vanish. The last financed project, $\sigma_a + (1-\rho_a^0)W(I) - m$, leaves the zone when ρ_a^0 or m rises, and the accelerator kept $\rho_a^0 W(I) + m$ on it. With the integrals running over the financed shocks and the density taken at the last one,

$$\text{(a)} \quad \int \frac{\partial}{\partial\rho_a^0}\big[(1-\rho_s^*)W(x^*)\big]g(\ell)\,d\ell = \big[\rho_a^0 W(I) + m\big]\,g\big(\sigma_a + (1-\rho_a^0)W(I) - m\big)\,W(I),$$

$$\text{(b)} \quad \int \frac{\partial}{\partial m}\big[(1-\rho_s^*)W(x^*)\big]g(\ell)\,d\ell = \big[\rho_a^0 W(I) + m\big]\,g\big(\sigma_a + (1-\rho_a^0)W(I) - m\big).$$

Combining (a) and (b), the boundary term disappears:

$$\text{(c)} \quad \int \Big[W(I)\frac{\partial}{\partial m} - \frac{\partial}{\partial\rho_a^0}\Big]\big[(1-\rho_s^*)W(x^*)\big]g(\ell)\,d\ell = 0.$$

(c) concerns the direction that lowers ρ_a^0 and raises m while leaving the last financed project in place. On a financed shock, its integrand equals

$$\frac{\sigma_s^* W(x^*)}{(\sigma_s^*+m)^2}\Big[(1-\rho_a^0)W(I) - m - \sigma_s^*\Big] - \Big[F(x^*) + \frac{\partial\Pi_s}{\partial\sigma_s}\Big]\Big[W(I)\frac{\partial\sigma_s^*}{\partial m} - \frac{\partial\sigma_s^*}{\partial\rho_a^0}\Big].$$

The first term is the gain from discrimination: at a given zone, the MFN component takes the more, the further the round is from the last financed one, at $\ell - \sigma_a = (1-\rho_a^0)W(I) - m$. On a shock without overinvestment, $\sigma_s^* = \ell - \sigma_a$ is below this edge, the first term is strictly positive and the second is zero. The second term is the cost on the shocks with overinvestment, where the move changes the seed investor's amount, and each extra dollar is lost on the projects that are not fully funded. □

Proof of Proposition 11

Against a priced SAFE of valuation σ_s/ρ_s, the seed investor's share is at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ (proof of Proposition 8). On a financed project, with the seed investor's amount σ_s^p, threshold x^p and profit Π_s^p, the accelerator keeps

$$W(I) - (\ell - \sigma_a) - \Pi_s^p - \int_{x^p}^{I} F(t)\,dt,$$

since $W(x^p) = W(I) + \sigma_s^p - (\ell-\sigma_a) - \int_{x^p}^{I} F(t)\,dt$. Its share never falls below ρ_a^0, so it keeps at least $\rho_a^0 W(I)$, what it keeps with $m=0$.

Suppose $m^*=0$. Then $(\sigma_a^*, \rho_a^{0*})$, written (σ_a, ρ_a^0), is the optimum of Proposition 6, with $\sigma_a > 0$ by assumption, and the seed investor finances the rounds $0 < \ell - \sigma_a \le (1-\rho_a^0)W(I)$, where the accelerator keeps $\rho_a^0 W(I)$. We add a small $m \in (0, \sigma_a)$ and show that the profit strictly rises. Fix a small $\delta > 0$.

The seed investor still finances almost all of the zone. On the round $\ell - \sigma_a$, consider the contract that invests the need for the share $\rho_s = (1-\rho_a^0)(\ell-\sigma_a)/(\ell - \sigma_a + m)$, of valuation $(\ell-\sigma_a+m)/(1-\rho_a^0)$. Above this valuation, the seed investor takes ρ_s. Below, it takes $(1-\rho_a^0)\rho_s(\pi - I)/(\rho_s(\pi-I)+m)$, and, using $1-\rho_a^0-\rho_s = (1-\rho_a^0)m/(\ell-\sigma_a+m)$, it misses

$$\frac{\rho_s\, m}{\rho_s(\pi-I)+m}\Big[1 - \frac{(1-\rho_a^0)(\pi-I)}{\ell-\sigma_a+m}\Big],$$

which, multiplied by $\pi - I$, is at most m. Its profit with this contract is at least

$$\frac{(1-\rho_a^0)W(I)\,(\ell-\sigma_a)}{\ell-\sigma_a+m} - (\ell-\sigma_a) - m,$$

which is nonnegative if and only if $\sqrt{(1-\rho_a^0)W(I)(\ell-\sigma_a)} \ge \ell - \sigma_a + m$. The difference of the two sides is concave in $\ell-\sigma_a$; for $m \le (1-\rho_a^0)W(I)/4$, it is nonnegative at $\ell-\sigma_a = 4m^2/((1-\rho_a^0)W(I))$ and at $\ell-\sigma_a = (1-\rho_a^0)W(I) - 3m$, hence between them. The priced seed investor, which does at least as well, finances every round in between.

On the rounds $\ell - \sigma_a \ge \delta$, the value kept gains at least $(1-\rho_a^0)\,m\,W(I)/(\ell - \sigma_a + m)$. The seed investor's share is at most $(1-\rho_a^0)\sigma_s^p/(\sigma_s^p+m)$, so Π_s^p is at most this share times $W(x^p)$, minus σ_s^p. With the expression of $W(x^p)$ above and $\int_{x^p}^{I} F(t)\,dt \le F(I)(\sigma_s^p - (\ell-\sigma_a))$,

$$\Pi_s^p + \int_{x^p}^{I} F(t)\,dt \le \frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p+m}\big[W(I) + \sigma_s^p - (\ell-\sigma_a)\big] - \sigma_s^p \\ + \Big(1 - \frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p+m}\Big)F(I)\big(\sigma_s^p - (\ell-\sigma_a)\big).$$

Moreover $x^p \geq 0$: otherwise $W(x^p) = E(\pi) - x^p$ and, since $E(\pi) < I$, the profit would be negative. Hence $\sigma_s^p - (\ell - \sigma_a) \leq I$. The upper bound has derivative in σ_s^p

$$\frac{(1-\rho_a^0)\,m}{(\sigma_s^p+m)^2}\Big[W(I) + \big(1-F(I)\big)\big(\sigma_s^p - (\ell-\sigma_a)\big)\Big] - \Big(1 - \frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p+m}\Big)\big(1-F(I)\big),$$

which is negative on the rounds $\ell - \sigma_a \geq \delta$ as soon as $m < \rho_a^0\big(1-F(I)\big)\delta^2/\big((1-\rho_a^0)(W(I)+I)\big)$, since $\sigma_s^p \geq \delta$ and the last factor in parentheses is at least $\rho_a^0 > 0$. The bound is then maximal at $\sigma_s^p = \ell - \sigma_a$, and the value kept is at least

$$\rho_a^0 W(I) + \frac{(1-\rho_a^0)\,m\,W(I)}{\ell - \sigma_a + m}.$$

The balance. On the financed rounds with $\ell - \sigma_a < \delta$, the value kept does not fall. The rounds lost lie in two bands, of widths $4m^2/((1-\rho_a^0)W(I))$ and $3m$; since g is bounded and the value kept on each is at most $W(I)$, the loss divided by m remains bounded. The gain on the rounds $\ell - \sigma_a \geq \delta$, divided by m, tends as $m \to 0$ to

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)} \frac{g(\ell)}{\ell-\sigma_a}\,d\ell,$$

which tends to $+\infty$ as $\delta \to 0$, since $g(\sigma_a) > 0$. For δ and then m small enough, the profit strictly rises, which contradicts $m^* = 0$. □

Proof of Proposition 12

Fix $\ell > \sigma_a$ and a seed round with amount $\sigma_s \geq \ell - \sigma_a$, threshold $x = I + \ell - \sigma_a - \sigma_s$, and any contract and protection. The seed investor receives at most the equity, worth $W(x)$ in expectation, so its profit is at most $W(x) - \sigma_s$. This bound has derivative $-F(x) \leq 0$ in σ_s: it is largest at $\sigma_s = \ell - \sigma_a$, where $x = I$, and equals $W(I) - (\ell - \sigma_a)$ there. The seed investor earns a nonnegative profit only if $\ell - \sigma_a \leq W(I)$. By (A1), $W(I) \leq E(\pi) < I$. □

Proof of Proposition 13

With an MFN clause against an unpriced SAFE, the seed investor finances exactly the shocks with $\ell - \sigma_a + m \leq (1-\rho_a^0)W(I)$ (Proposition 8), and the accelerator keeps $\rho_a^0 W(I) + m$ on the last one (proof of Proposition 10). With $m = 0$, this is the zone and the share of the fixed share $\rho_a = \rho_a^0$ (Proposition 4); with $\rho_a^0 = m = 0$, the zone of Proposition 1, where the accelerator keeps nothing. In all cases, the accelerator covers $[0, \sigma_a]$ alone, and the experiment is covered up to $\sigma_a + (1-\rho_a^0)W(I) - m$. □

Proof of Proposition 14

(i) When the seed investor covers only its need, its threshold is I (Propositions 1 and 4). If it finances the shocks up to $\bar{\ell}$, the probability of full funding is

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell + \big(1 - F(I)\big)\big[G(\bar{\ell}) - G(\sigma_a)\big].$$

Its derivative in $\bar{\ell}$ is $(1 - F(I))\, g(\bar{\ell}) > 0$, since $F(I) < 1$, as $W(I) > 0$. Its derivative in σ_a, at a given $\bar{\ell}$, is $\int_0^{\sigma_a} f(I + \ell - \sigma_a)\, g(\ell)\, d\ell$, the terms $(1 - F(I))g(\sigma_a)$ of the two integrals canceling. With a fixed share ρ_a, the last shock is $\sigma_a + (1 - \rho_a)W(I)$, and moves one for one with σ_a: at a given ρ_a, the derivative in σ_a is

$$\int_0^{\sigma_a} f(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \big(1 - F(I)\big) g\big(\sigma_a + (1 - \rho_a)W(I)\big) > 0.$$

Without protection, $\rho_a = 0$. We go from $(\sigma_a^B, 0)$ to (σ_a^P, ρ_a^P) in two steps. Raising the contribution from σ_a^B to σ_a^P at $\rho_a = 0$ gives the integral of the derivative above, the first term. Raising the share from 0 to ρ_a^P at σ_a^P moves the last shock from $\sigma_a^P + W(I)$ down to $\sigma_a^P + (1 - \rho_a^P)W(I)$, and gives the second term, strictly negative since $\rho_a^P > 0$.

(ii) On a round where the seed investor invests $\sigma_s^* > \ell - \sigma_a$, the threshold is $x^* = I + \ell - \sigma_a - \sigma_s^* < I$. Since $f > 0$, $F(x^*) < F(I)$, and the project is fully funded with probability $1 - F(x^*) > 1 - F(I)$. □

Proof of Proposition 15

Analysis. By Proposition 13, the probability of a seed round is $G(\sigma_a + (1 - \rho_a^0)W(I) - m) - G(\sigma_a)$. Under the MFN clause, the probability of full funding is

$$\int_0^{\sigma_a^M} \big(1 - F(I + \ell - \sigma_a^M)\big) g(\ell)\, d\ell + \big(1 - F(I)\big)\Big[G\big(\sigma_a^M + (1 - \rho_a^{0M})W(I) - m^M\big) - G(\sigma_a^M)\Big] + \int_{\sigma_a^M}^{\sigma_a^M + (1 - \rho_a^{0M})W(I) - m^M} \big(F(I) - F(x^*(\ell))\big) g(\ell)\, d\ell,$$

and under the fixed share, where the seed investor invests its need (Proposition 4), the same expression with $(\sigma_a^P, \rho_a^P, 0)$ and without the last integral.

Synthesis. *The probability of a seed round.* We add and subtract $G(\sigma_a^M + (1 - \rho_a^P)W(I)) - G(\sigma_a^M)$ in the difference, which gives the stated sum.

The probability of full funding. We go from the fixed share to the MFN clause in two steps. Raising the contribution from σ_a^P to σ_a^M at the share ρ_a^P gives the integral of the derivative in the proof of Proposition 14, the first term. Moving the last shock from $\sigma_a^M + (1 - \rho_a^P)W(I)$ to $\sigma_a^M + (1 - \rho_a^{0M})W(I) - m^M$ gives the second. The last integral is nonnegative, since $x^*(\ell) \leq I$ on every round. □

Proof of Proposition 16

Write $\bar{\ell} = \sigma_a + (1-\rho_a^0)W(I) - m$ for the last shock the seed investor finances.

(i) On a shock ℓ, the seed investor invests σ_s^*, the threshold is $x^* = I + \ell - \sigma_a - \sigma_s^*$, and the accelerator keeps $(1-\rho_s^*)W(x^*) = W(x^*) - \sigma_s^* - \Pi_s^*(\ell)$. Since $W' = -(1-F)$ and $I - x^* = \sigma_s^* - (\ell - \sigma_a)$,

$$W(x^*) - \sigma_s^* = W(I) + \int_{x^*}^{I} \big(1 - F(x)\big)dx - \sigma_s^* = W(I) - (\ell - \sigma_a) - \int_{x^*}^{I} F(x)\,dx.$$

By Proposition 13, the seed investor finances exactly $(\sigma_a, \bar{\ell}]$, and the accelerator covers $[0, \sigma_a]$ alone; adding $B(\sigma_a)$ gives the profit.

(ii) At $\bar{\ell}$, the seed investor earns zero and invests its need (proof of Proposition 10), so the integrand of (i) equals $W(I) - (\bar{\ell} - \sigma_a) = \rho_a^0 W(I) + m$ there, and $\bar{\ell}$ moves by $-W(I)$ per unit of ρ_a^0. Differentiating (i),

$$\frac{\partial \Pi_a}{\partial \rho_a^0} = -\int_{\sigma_a}^{\bar{\ell}} \frac{\partial}{\partial \rho_a^0}\Big(\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\,dx\Big) g(\ell)\,d\ell - W(I)\big(\rho_a^0 W(I) + m\big) g(\bar{\ell}).$$

The optimal fixed share ρ_a at σ_a with the same last shock keeps $\rho_a W(I) = W(I) - (\bar{\ell} - \sigma_a) = \rho_a^0 W(I) + m$ on it, and satisfies $G(\bar{\ell}) - G(\sigma_a) = \rho_a W(I)\, g(\bar{\ell})$ (Proposition 6). Hence $W(I)(\rho_a^0 W(I) + m)\, g(\bar{\ell}) = \int_{\sigma_a}^{\bar{\ell}} W(I)\, g(\ell)\,d\ell$, and

$$\frac{\partial \Pi_a}{\partial \rho_a^0} = -\int_{\sigma_a}^{\bar{\ell}} \Big[W(I) + \frac{\partial}{\partial \rho_a^0}\Big(\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\,dx\Big)\Big] g(\ell)\,d\ell.$$

Lowering ρ_a^0 extends the zone, and pays if and only if this derivative is negative. On a round without investment beyond the need, $x^* = I$, the lost surplus is zero, and $\partial \Pi_s^*/\partial \rho_a^0 = -(\ell - \sigma_a)W(I)/(\ell - \sigma_a + m)$; the integrand equals $mW(I)/(\ell - \sigma_a + m)$. On a round with investment beyond the need, the lost surplus has derivative $F(x^*)\,\partial \sigma_s^*/\partial \rho_a^0$, since x^* falls one for one with σ_s^*, and $\partial \sigma_s^*/\partial \rho_a^0 < 0$ by the proof of Proposition 10.

(iii) We differentiate (i) in σ_a, with ρ_a^0 and m fixed. The bracket of (i) is $W(I) - (\ell - \sigma_a) - \varphi(\ell - \sigma_a)$, where φ, the seed investor's profit plus the lost surplus, depends only on the round. The width of the zone, $(1-\rho_a^0)W(I) - m$, does not depend on σ_a. Changing variable to the round $t = \ell - \sigma_a$, the integral of (i) is $\int_0^{(1-\rho_a^0)W(I)-m} [W(I) - t - \varphi(t)] g(\sigma_a + t)\,dt$, with derivative

$$\int_0^{(1-\rho_a^0)W(I)-m} \big(W(I) - t - \varphi(t)\big) g'(\sigma_a + t)\,dt.$$

An integration by parts gives

$$\int_0^{(1-\rho_a^0)W(I)-m} (W(I) - t)\, g'(\sigma_a + t)\,dt = \big(\rho_a^0 W(I) + m\big) g(\bar{\ell}) - W(I)\, g(\sigma_a) + G(\bar{\ell}) - G(\sigma_a).$$

Adding $B'(\sigma_a) = W(I)g(\sigma_a) - (1 - G(\sigma_a)) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)g(\ell)\,d\ell$ gives the stated derivative.

With a fixed share, $\varphi(t) = (1 - \rho_a)W(I) - t$. An integration by parts gives

$$-\int_0^{(1-\rho_a)W(I)} \varphi(t)\, g'(\sigma_a + t)\,dt = (1 - \rho_a)W(I)g(\sigma_a) - \big[G(\bar{\ell}) - G(\sigma_a)\big],$$

and the derivative reduces to the one in the proof of Proposition 6. □

Proof of Proposition 17

We show that the total surplus of each covered project is $W(I) - \ell - \int_x^I F(t)\,dt$, then sum over the shocks.

Analysis. On a covered project with threshold x, the accelerator and the seed investor bring $I + \ell - x$ before the equity round, and the equity investor brings x for an equity worth x in expectation. The payoffs of the three agents add up to $W(x) - (I + \ell - x)$. Since $W' = -(1 - F)$,

$$W(x) = W(I) + \int_x^I \big(1 - F(t)\big)\,dt = W(I) + (I - x) - \int_x^I F(t)\,dt,$$

and the total surplus of the project is $W(I) - \ell - \int_x^I F(t)\,dt$. An uncovered project is abandoned, and its surplus is $-\sigma_a$, the contribution spent.

Synthesis. The accelerator covers $[0, \sigma_a]$ alone at the threshold $I + \ell - \sigma_a$, and the seed investor finances $(\sigma_a, \bar{\ell}]$ at the threshold $x^*(\ell)$ (Proposition 13). Summing,

$$TS = \int_0^{\bar{\ell}} \big(W(I) - \ell\big)\, g(\ell)\,d\ell - L_{acc} - L_{over} - \sigma_a\big(1 - G(\bar{\ell})\big),$$

and the first integral is $TS^* - L_{cov}$. If $\bar{\ell} < W(I)$, L_{cov} integrates the positive $W(I) - \ell$ over $(\bar{\ell}, W(I))$; if $\bar{\ell} > W(I)$, it equals $\int_{W(I)}^{\bar{\ell}} (\ell - W(I))g(\ell)\,d\ell \geq 0$.

The derivatives. When the seed investor invests its need, $L_{over} = 0$, and the total surplus depends on the contribution and on the last shock:

$$\int_0^{\bar{\ell}} \big(W(I) - \ell\big)\, g(\ell)\,d\ell - L_{acc}(\sigma_a) - \sigma_a\big(1 - G(\bar{\ell})\big).$$

Its derivative in $\bar{\ell}$ is $(W(I) - \bar{\ell} + \sigma_a)\, g(\bar{\ell})$: the value of the last project at the threshold I net of its round, against $-\sigma_a$ if it were abandoned. With $\bar{\ell} = \sigma_a + W(I) - (\rho_a^0 W(I) + m)$, a dollar more kept on the last project lowers $\bar{\ell}$ by one dollar, and changes the total surplus by $-(\rho_a^0 W(I) + m)\, g(\bar{\ell})$. A dollar more of contribution, at a given amount kept, raises $\bar{\ell}$ by one dollar: the total surplus changes by $(\rho_a^0 W(I) + m)g(\bar{\ell})$ through the last shock, by $-(1 - G(\bar{\ell}))$ through the last term, and by $-dL_{acc}/d\sigma_a = -\int_0^{\sigma_a} F(I + \ell - \sigma_a)g(\ell)\,d\ell$ through L_{acc}, by Leibniz, the boundary term vanishing at $\ell = \sigma_a$. □

Proof of Proposition 18

Without protection and with the fixed share, the seed investor invests its need (Propositions 1 and 4). We go from $(\sigma_a^B, 0)$ to (σ_a^P, ρ_a^P) in two steps. Raising the contribution from σ_a^B to σ_a^P without protection, where the accelerator keeps nothing on the last project, gives the integral of $-(1 - G(s + W(I))) - \int_0^s F(I + \ell - s)g(\ell)\, d\ell$, the first term. Raising the share from 0 to ρ_a^P at σ_a^P moves the last shock from $\sigma_a^P + W(I)$ down to $\sigma_a^P + (1-\rho_a^P)W(I)$, and loses on each shock in between its value $\sigma_a^P + W(I) - \ell$ against abandonment, the second term, strictly negative since $\rho_a^P > 0$. □

Proof of Proposition 19

With the MFN clause, the total surplus is the expression of the proof of Proposition 17 minus L_{over}, and with the fixed share the same expression with $L_{over} = 0$. We go from the fixed share to the MFN clause in two steps. Raising the contribution from σ_a^P to σ_a^M at the share ρ_a^P gives the integral of the derivative in the contribution of the proof of Proposition 17, with $\rho_a^0 W(I) + m = \rho_a^P W(I)$, the first term. Moving the last shock from $\sigma_a^M + (1 - \rho_a^P)W(I)$ to $\sigma_a^M + (1 - \rho_a^{0M})W(I) - m^M$ adds or removes, on each shock in between, its value $\sigma_a^M + W(I) - \ell$ against abandonment, the second term. L_{over} is nonnegative since $x^*(\ell) \leq I$ on every round. □

Proof of Proposition 20

The seed investor. The seed investor's profit depends on the protections only through the share they leave it. The founders' stake is written before the seed round and comes first, like the accelerator's protection: by the rule of Section 2, with a fixed share ρ_a the seed investor's share is at most $1 - \rho_e - \rho_a$, and with an MFN clause the SAFEs signed after the protections share $1 - \rho_e - \rho_a^0$. The seed investor's program is therefore the program of the corresponding proposition with ρ_a replaced by $\rho_e + \rho_a$, or ρ_a^0 by $\rho_e + \rho_a^0$. Without protection, it is the program of Propositions 4 and 5 with $\rho_a = \rho_e$: the seed investor invests its need, takes $1 - \rho_e$, finances the shock if and only if $\ell - \sigma_a \leq (1 - \rho_e)W(I)$, and the cap $(\ell - \sigma_a)/(1 - \rho_e) = \sigma_s^*/(1 - \rho_e)$ gives it this share in every state above I.

The total surplus. With $\sigma_a = 0$ and no protection, the covered projects are the shocks $\ell \leq (1 - \rho_e)W(I)$, at the threshold I, with surplus $W(I) - \ell$, and the other projects are abandoned with surplus 0. Hence

$$TS = \int_0^{(1-\rho_e)W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell = TS^* - \int_{(1-\rho_e)W(I)}^{W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell,$$

and the last integral is positive since $g > 0$ and $\rho_e > 0$. □